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Early solar system timescales according to ^{53}Mn-^{53}Cr systematics

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Abstract—We present results of a study of the ^{53}Mn-^{53}Cr systematics in various solar system objects: angrites, eucrites, chondrites, diogenites, pallasites, the Earth and the Moon, and SNC meteorites. The primary goal of this study was to explore the capabilities of the ^{53}Mn-^{53}Cr isotope system as a chronometer and as a tracer for events in the early solar system, to obtain chronological information on different classes of meteorites, and to investigate the indigenous distribution of ^{53}Mn in the late nebula.

These studies have shown that all meteorite groups investigated so far have excess ^{53}Cr relative to the terrestrial value. A lunar sample exhibits ^{53}Cr/^{52}Cr ratios which are the same as the terrestrial normal. The angrites, several eucrites, and the pallasites show clear evidence for the existence of life ^{53}Mn during their formation while other meteorites were isotopically equilibrated after essentially all ^{53}Mn had decayed. A well defined whole-rock ^{53}Mn-^{53}Cr isochron for the HED (Howardite-Eucrite-Diogenite) parent body was obtained. The isochron indicates that this planetesimal was essentially totally molten and differentiated ~7 Ma before the angrites crystallized. Using the absolute age of the angrites as a time marker this event has occurred 4565 Ma ago, within present uncertainties at the same time when high temperature meteorite inclusions (CAI) were formed in the nebula. The first basalts were deposited onto its surface within less than 3 Ma. The bulk Mn/Cr ratios of the HED parent body (presumably Vesta), the angrites, and the pallasites are consistent with a chondritic Mn/Cr ratio.

The results from the SCN meteorites show that their ^{53}Cr excesses are less than half of those found in the other meteorites. Thus, the characteristic ^{53}Cr/^{52}Cr ratio of Mars (assuming SNCs originate from this planet) are intermediate between that of the earth-moon system and those of the other meteorites. When these ^{53}Cr excesses are plotted as a function of the heliocentric distance of the place of origin of the samples then a linear relationship is indicated. Provided that this variation is due to the decay of ^{53}Mn then a radial heterogeneous distribution of ^{53}Mn must have existed in at least the inner early solar system. It is argued that radial fractionation within the nebula, based on the slightly higher volatility of Mn as compared to that of Cr, is an unlikely cause for this distribution. Thus, it must be an intrinsic feature of the late solar nebula. Stochastic mixing processes at the planetary embryo stage did obviously not eradicate this heterogeneity.

Based on the ^{53}Mn-^{53}Cr systematics in HED meteorites and in chondrites rather narrow limits are inferred for the age of the solar system or, more accurately, for the start of the decay of ^{53}Mn within the solar nebula; the range of possible values is 4568–4571 Ma. The lower limit is consistent with the Pb-Pb ages of CAI's.

1. INTRODUCTION

The former presence in ancient solar system objects of short-lived radionuclides with half lives of 10^5 to 10^8 years may either be due to nucleosynthetic production within stars and subsequent rapid transport to and injection into the nascent solar system or, possibly, local production by highly energetic particles from the sun. These radionuclides, although now extinct, were still alive at the time of formation of ancient solids. The decay of these short-lived species is now revealed in the form of anomalies in the isotopic composition of the daughter elements. Since the time of the first proposals to look for such effects (Brown, 1947; Süss, 1948) the evidence for the presence of almost a dozen of extinct radionuclides in the solar system was found (see the recent review by Podosek and Nichols, 1997). The importance of these now extinct short-lived radionuclides is that they can provide constraints for models of nucleosynthesis and may serve as sensitive chronometers and tracers for processes in the early solar system.

One of these radionuclides, ^{53}Mn, with a half life of 3.7 ± 0.4 Ma (Honda and Imamura, 1971), is especially suitable for high resolution chronological studies for the first ~20 Ma of solar system history. The former presence of ^{53}Mn is indicated by excesses relative to the terrestrial standard value of the daughter nucleus, ^{53}Cr. One of the advantages of the ^{53}Mn-^{53}Cr chronometer is that Mn and Cr are abundant elements in solar-system materials and that the ^{53}Mn-^{53}Cr system, therefore, is potentially useful for dating a wide variety of ancient objects. Another advantage is the presence in many objects of a chromite-spinel phase which has a very low Mn/Cr ratio. This permits to obtain initial ^{53}Cr/^{52}Cr ratios at the time of isotope closure of the Mn-Cr system with a high precision.

After the pioneering finding of the vestiges of ^{53}Mn in the Allende refractory inclusions (CAI's) (Birck and Allègre, 1985a), excesses of ^{53}Cr were detected in various solar system objects: carbonaceous chondrites (Birck and Allègre, 1988; Birck et al., 1990; Rotaru et al., 1989; Rotaru et al., 1990; Harper and Wiesmann, 1992; Rotaru et al., 1992; Endress et al., 1996), enstatite chondrites (Birck and Allègre, 1988; Wadhwa et al., 1997a), ordinary chondrites (Lugmair and McIsaac, 1995; Nyquist et al., 1997), pallasites (Birck and Allègre, 1988; Hutcheon and Olsen, 1991; Hsu et al., 1997; Shukolyukov and Lugmair, 1997a), iron meteorites (Hutcheon et al., 1985; Davis and Olsen, 1990; Hutcheon and Olsen, 1991; Hutcheon et al., 1992), angrites (Nyquist et al., 1991, 1994; Lugmair et al., 1992); eucrites (Lugmair et al., 1992 and 1994a,b; Nyquist et al., 1996; Lugmair and Shukolyukov, 1997; Shukolyukov and Lugmair, 1997b), mesosiderites (Wadhwa et. al., 1997b), SNC meteorites (Lugmair et al., 1996), and primitive achondrites Acapulco (Harper et al., 1992; Zipfel et al., 1996) and Divnoe (Bogdanovski et al., 1997). In many cases the measured ^{53}Cr excesses in mineral phases of these meteorites are correlated with the respective Mn/Cr ratios, which indicates in situ decay of ^{53}Mn and allows calculation from the slope of the correlation line, a relative abundance of ^{53}Mn (that is the ^{53}Mn/^{55}Mn ratio) at the time of isotope closure. The relative ages of two meteorites, 1 and 2, are then calculated from their ^{53}Mn/^{55}Mn ratios:

$$\Delta T_{1\text{-}2} = 1/\lambda \ln[(^{53}Mn/^{55}Mn)_1/(^{53}Mn/^{55}Mn)_2] \qquad (1)$$

where λ is the decay constant. Provided that a sufficiently precise absolute age for one of the meteorites is known, relative ^{53}Mn-^{53}Cr ages can then be converted into absolute ages (section 3.1.1). There is, however, an important prerequisite for using the ^{53}Mn-^{53}Cr isotope system as a chronometer. The ^{53}Mn/^{55}Mn ratios obtained for meteorites from different parent bodies bear chronological meaning only if the ^{53}Mn/^{55}Mn ratios in the accretion regions where these parent bodies formed were identical at the same time. Thus, before assigning a meaningful relative age to meteorites from different parent bodies it is crucial to examine closely the assumption of homogeneous spatial distribution of ^{53}Mn in this region or in the solar nebula as a whole.

The goal of this paper is not to review all ^{53}Mn-^{53}Cr data available in the literature although, in section 3.4, we will discuss earlier Mn-Cr results as they relate to our data. We summarize and focus on the results obtained in our laboratory during the last several years and present our current view on ^{53}Mn-^{53}Cr systematics in the solar system.

We initiated our study of the ^{53}Mn-^{53}Cr isotope system in 1990 with the angrite LEW86010 with the aim of mapping the relative ^{53}Mn-^{53}Cr chronometer onto an absolute time scale. Since then results for eucrites, diogenites, ordinary chondrites, pallasites, primitive achondrites, SNC meteorites, and the Moon have been obtained. Some data, as they were published in preliminary form during earlier years, were refined during the course of our study due to improvements in our analytical procedures. In an attempt to make the logic behind our investigations more transparent, we have tried to present our results (Section 3) by following a sequential approach and to discuss the issues in the same order as the respective questions have arisen.

2. EXPERIMENTAL PROCEDURES

2.1. Sample Preparation and Chemical Procedures

Regardless of subsequent treatments (dissolution of bulk samples, differential dissolution of minerals, physical separation of mineral fractions) we followed several general procedures for most of the samples. To minimize potential surface contamination the fusion crust and severely weathered zones were avoided. When possible, samples were extracted from the inner sections of meteorites. Prior to dissolution, all samples were thoroughly washed in water, acetone, and methanol under ultrasonication. For the extraction of samples clean tungsten tools were used. Samples were crushed in a boron carbide mortar. To ensure better representation for the analyses of whole rocks, reasonably large bulk samples were analyzed. Usually, when a sufficient amount of material was available, several hundred mg samples were dissolved and a 5–30 % split of this sample solution was then processed through chemistry. In cases when weathered areas could not be avoided, additional washing in dilute HCl or HNO_3 was applied to remove potential terrestrial contamination and alteration products. Mineral separates were obtained with a clean Frantz Isodynamic magnetic separator followed by hand-picking. In some cases, e.g., for generating pyroxene and chromite fractions from Serra de Magé, only hand-picking was utilized. Mineral separates were also washed in water, acetone, methanol, and dilute HCl or (and) HNO_3. We have found that in all cases the amounts of both Mn and Cr which were removed during washing were negligible.

Washing, dissolution, and all other chemical procedures were performed under clean lab conditions. After washing, dried bulk samples and mineral separates were weighed and dissolved in an HF/HNO_3 mixture under ultrasonification followed with HCl. Olivine separates were dissolved directly in 6–8 N HCl. The olivine solutions were subsequently dried and the precipitates were attacked with an HF/HNO_3 mixture to decompose the remaining silica gel with eventual dissolution in HCl. To dissolve chromite (spinel) fractions we applied an HF/HNO_3 mixture at elevated temperature (170–180°C) using a teflon pressure bomb. The high resistance of chromite to dissolution in acids allows easy separation of this phase from its silicate counterpart by a simple treatment of the original sample with an HF/HNO_3 mixture at room temperature. Metal samples were dissolved directly in 4.5 N HCl. To obtain silicate and chromite fractions of bulk chondrites the material was initially treated with 4.5 N HCl to remove the metal phase followed by the dissolution of silicates in an HF/HNO_3 mixture.

After dissolution and equilibration for at least an hour, small aliquots of the solutions were taken for the measurement of Cr and Mn concentrations. The concentration measurements were made by graphite furnace atomic absorption (AA) spectrometry using standard addition techniques. Iron concentrations (when applicable) were determined by flame AA. From reproducibility and comparison to standard samples the accuracy of the concentration measurements is estimated to be $\sim\pm5\%$.

The separation of Cr was similar to that of Birck and Allègre (1988). It is based on the differences in ion exchange properties of various hydrated species of Cr^{+3}. The separations were performed using cation exchange resin Mitsubishi AG$\times$8 (200–400 mesh) and included two successive elutions: with 1 N HCl in a primary column to isolate Cr from major matrix elements and with 0.5 N HF, 1 N HCl, and 1.8 N HCl in an additional clean-up column to remove remaining traces of Fe, alkalies, and anions. The total yield of Cr in the chemical procedures was ~70%. Chromium blanks were on the order of several ng and are totally insignificant.

2.2. Mass Spectrometry

The isotopic analyses of Cr were made with a VG-54E thermal ionization single-collector mass spectrometer. Chromium was loaded in dilute HCl on single prebaked W filaments and was analyzed as Cr^+. A mixture of silica gel and boric acid was used as an emitter. The Cr amount per load was usually 600–700 ng. The loading procedure was refined since our early work (Lugmair et al., 1992), and loading conditions such as amounts of silica gel and boric acid, drying procedure, sample area on a filament, etc., were standardized. Ionization efficiencies ranged from 2 to 5×10^{-3}. The running temperature was typically 1150–1190°C and the ion-current for ^{52}Cr of 5.5×10^{-11}A was held constant to within $\pm10\%$. For each sample, isobaric interferences were monitored via background scans before and sometimes during the experiment with the use of a Daly detector. Interferences from ^{50}Ti, ^{50}V, and ^{54}Fe were monitored at ^{48}Ti, ^{51}V, and ^{56}Fe and were always negligible. A chromium shelf standard solution and a chemically processed geological sample (San Carlos olivine) were used as a terrestrial reference material and test samples. All chromium isotopes were normalized to ^{52}Cr and corrected for mass fractionation

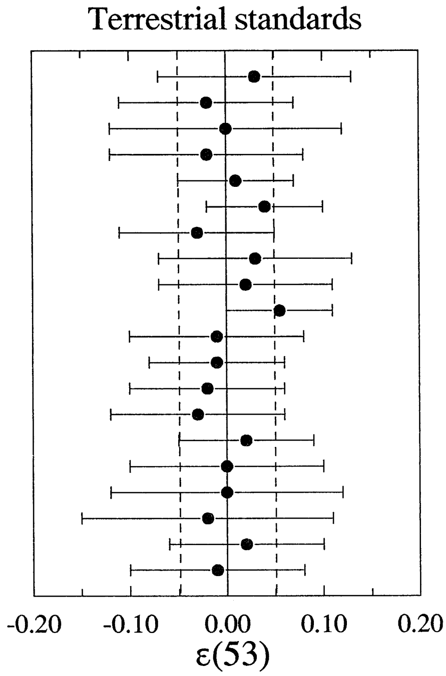

Fig. 1. $^{53}Cr/^{52}Cr$ in terrestrial standards. The $^{53}Cr/^{52}Cr$ ratios are expressed as the deviations from the overall average standard $^{53}Cr/^{52}Cr$ in ε-units ($\varepsilon \equiv [(^{53}Cr/^{52}Cr)_{measured}/(^{53}Cr/^{52}Cr)_{average}$ - 1] $\times 10^4$). Each data point represents the average $^{53}Cr/^{52}Cr$ of 16 standard measurements. The $^{53}Cr/^{52}Cr$ ratios for single measurements are corrected for second order fractionation effects by using the $^{53}Cr/^{52}Cr$ vs.$^{54}Cr/^{52}Cr$ correlation (see text). The presented uncertainties are based on estimates from the reproducibility of the repeat measurements (external precision); the $2\sigma_{mean}$ uncertainties would be a factor of two smaller. The total spread of the data points, which is shown by the dashed lines, is $\pm$5 ppm. The standard value of $^{53}Cr/^{52}Cr$ is taken as 0.1134507 with a concomitant $^{54}Cr/^{52}Cr$ ratio of 0.0282085.

according to an exponential law with the use of $^{50}Cr/^{52}Cr = 0.051859$ (Shields et al., 1966) as internal standard. Instrumental fractionation for normal sample loads ranged typically from -7 to -1.8 permil per amu using this value. We observed, however, the presence of a residual mass fractionation effect from the application of a strict exponential law. After exponential correction we found a strongly linearly correlated variation (R $\sim$ 0.99, slope = 2.15) of these residual effects on $^{53}Cr/^{52}Cr$ vs. $^{54}Cr/^{52}Cr$ for the laboratory standards. These residual effects were independent of instrumental fractionation. They seem to only depend on the actual location of sample evaporation on the filament (relative to the ion-source optics) and on the evaporation/diffusion process itself. When determining $^{53}Cr/^{52}Cr$ ratios for samples, these small mass discrimination effects were corrected by utilizing the parameters (slope and intercept) obtained from the $^{53}Cr/^{52}Cr$ vs $^{54}Cr/^{52}Cr$ correlation line from more than 200 Cr standard measurements. The uncertainties on the correlation parameters were not propagated. For each sample typically 16–20, but occasionally up to thirty repeat measurements of the chromium isotopic composition were performed, and the results were averaged. A single measurement or run consisted of thirty blocks of ten ratios each (i.e., cycles through all mass peaks and baseline) with peak integration times of 3–5 sec depending on peak intensity, and typical $2\sigma_{mean}$ uncertainties on $^{53}Cr/^{52}Cr$ of 0.1–0.2 ϵ; normally four runs were performed per sample load over night. The errors of the results from a single sample load did not show any correlation. Figure 1 documents the reproducibility of Cr isotopic analyses. Each data point represents the average $^{53}Cr/^{52}Cr$ (corrected using the $^{53}Cr/^{52}Cr$ vs $^{54}Cr/^{52}Cr$ correlation) of sixteen measurements of the laboratory standard; that is, to construct the graph, we treated our standard measurements the same way as we treated sample measurements. The $^{53}Cr/^{52}Cr$ ratios are expressed as the deviations from the overall average standard $^{53}Cr/^{52}Cr$ in ϵ-units ($\epsilon \equiv [(^{53}Cr/^{52}Cr)_{measured}/(^{53}Cr/^{52}Cr)_{average}$ - 1] $\times 10^4$). The total spread of the data points is only $\pm$5 ppm. The presented uncertainties are based on the 1σ distribution of the individual run results and are used to reflect external intermediate term reproducibility. Although normally one would use the $2\sigma_{mean}$ uncertainties, we were concerned that these smaller errors would not adequately take into account unknown systematic effects (such as machine bias drift etc.) which can modulate with periods on the order of months but cannot be sufficiently resolved with only 10–20 standard runs during that time period. The $^{53}Cr/^{52}Cr$ ratios for samples are also expressed in ϵ-units: $\epsilon \equiv [(^{53}Cr/^{52}Cr)_{sample}/(^{53}Cr/^{52}Cr)_{standard}$ - 1] $\times$ 10^4. The standard value for $^{53}Cr/^{52}Cr$ is taken as 0.1134507. As mentioned above, $(^{53}Cr/^{52}Cr)_{sample}$ for each sample measurement was calculated from the $^{53}Cr/^{52}Cr$ vs. $^{54}Cr/^{52}Cr$ correlation line. This second-order fractionation correction enabled us to considerably improve the reproducibility and precision of the chromium isotopic composition measurements. Obviously, the disadvantage of this procedure is that it cannot be used for samples with an anomalous $^{54}Cr/^{52}Cr$ ratio, such as Allende CAI's, which contain a presolar Cr component with nuclear isotopic anomalies. However, such samples were not studied here. Thus, for all samples studied in this work the $^{54}Cr/^{52}Cr$ ratios, corrected for exponential fractionation in the usual manner, are normal within $<\pm 0.2$ ϵ, and there is no evidence for $^{54}Cr/^{52}Cr$ to be different from the terrestrial value.

Figure 1 shows that the total spread of the $^{53}Cr/^{52}Cr$ ratios for the standards (expressed in blocks of sixteem measurements) significantly smaller than the listed uncertainties (normally $\pm$0.07–0.10 ϵ). This may suggest that the errors we use may be too conservative. This is also apparent for samples and is illustrated by the results for the meteorites with equilibrated Cr isotopic composition, e.g., for the Caldera samples (section 3.1.2). It has to be emphasized, however, that the goal of performing 16–30 repeat measurements of the chromium isotopic composition for a single sample was not primarily to decrease the error but to obtain the best value possible.

2.3. Spallation Correction

Since the measured excesses of radiogenic ^{53}Cr are very small (see below) for the samples with relatively low Cr concentrations, such as olivines and those with long exposure ages, a potential contribution from spallation reactions has to be taken into account. Using the $^{54}Cr/^{52}Cr$ ratio for a second order fractionation correction (see above) introduces an additional uncertainty due to the presence of spallogenic ^{54}Cr.

Birck and Allègre (1985b) determined production rates for several elements, including Cr, in the interior of the iron meteorite Grant. The production rates for ^{53}Cr and ^{54}Cr turned out to be approximately the same: $\sim 2.9 \times 10^{11}$ atoms/g per Ma (Birck and Allègre, 1985; Birck, pers. commun.). This value agrees reasonably well with the less precise values for the production rate of ^{53}Cr in Grant of $\sim 2.3 \times 10^{11}$ atoms/g per Ma by Shimamura et al. (1986) and $\sim 2.7 \times 10^{11}$ atoms/g per Ma by Shima and Honda (1966). Thus, with known exposure age and Cr and Fe concentrations (Fe is the main target for Cr production), a spallation correction can be performed.

In some cases, however, exposure ages are not known, or, as for the pallasites Omolon and Springwater, are poorly constrained. Furthermore, both shielding conditions and depth dependence of the Cr production rates are not known which may introduce an additional error.

For this reason we have evaluated the contribution of the spallation component in the pallasite olivines by direct measurement of the chromium isotopic composition in the metal phase. If exposure ages are long, the spallation ^{53}Cr in metal is predominant as compared to the radiogenic ^{53}Cr. The concentration of the indigenous Cr in metal is low. Thus, the excess of ^{53}Cr is almost entirely due to spallation and is readily measurable. Since the only target for the production of spallogenic Cr in olivines is Fe, the isotopic composition of spallation Cr should be identical in both phases (provided that samples are taken from neighboring locations within a meteorite). Obviously, the magnitude of the spallation contribution is proportional to the Fe/Cr ratio. Therefore, using data on the spallation Cr in metal, the spallation correction in olivines can be accomplished by comparing the Fe/Cr ratios in metal and in olivine. The concentration of Cr in the Omolon

Table 1. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in angrites

Sample*	Weight (mg)	Mn (ppm)	Cr(ppm)	^{55}Mn/^{52}Cr ($\pm$5%)	$\epsilon(53)^{\dagger}$
LEW TR	56	1530	755	2.29	0.66 ± 0.31
LEW Px−1	31	703	1268	0.63	0.48 ± 0.29
LEW Px−2	4	696	1610	0.49	0.45 ± 0.25
LEW Ol	44	2947	20.8	160	$18.07 \pm 0.48^{\ddagger}$
ADOR Px	22	880	1826	0.54	0.44 ± 0.12
ADOR Ol	14	3557	33.7	$119^{\S}$	$11.9 \pm 4.2^{\ddagger}$

* LEW 86010 (LEW), Angra dos Reis (ADOR), total rock (TR), pyroxene (Px), olivine (Ol).

† Here and in the following tables $\epsilon(53) \equiv [(^{53}Cr/^{52}Cr)_{sample}/(^{53}Cr/^{52}Cr)_{standard} - 1] \times 10^4$; the uncertainties on $\epsilon(53)$ are based on estimates of the reproducibility of the repeat measurements of the sample and standards (see section 2.2).

‡ Only two (for ADOR Ol at low intensity) runs for the Cr isotopic composition could be made because of limited sample amounts.

§Uncertainty on the ^{55}Mn/^{52}Cr ratio is 10%.

metal, however, turned out to be too small (less than 0.3 ppm) which was below the measurable limit for the sample used. For this reason a known amount of standard Cr (which served as a carrier for the spallogenic Cr) was added to the metal solution and the chromium isotopic composition of the mixture was measured. A final correction for the spallation component on ^{53}Cr/^{52}Cr in the Omolon olivines 1 and 2 (including both the direct ^{53}Cr correction and the correction of ^{53}Cr via ^{54}Cr) was found to be 0.14 ± 0.01 ϵ and 0.12 ± 0.01 ϵ, respectively. This demonstrates that the applied procedure provides an accurate correction for the spallation component without the knowledge of an exposure age, shielding conditions, and Cr production rates.

3. RESULTS AND DISCUSSION

3.1. Chronology of Angrites and Noncumulate Eucrites

3.1.1. Angrites as absolute time markers

The main aim for studying the angrites was to tie the relative ^{53}Mn-^{53}Cr chronometer to an absolute time scale (Lugmair et al., 1992). We considered the angrites LEW86010 (LEW) and Angra dos Reis (ADOR) as ideal samples for this purpose. Firstly, the angrites are early equilibrated planetary differentiates which cooled fast, do not show any signs of later disturbance, and their ages are precisely known. The Pb-Pb ages of both meteorites of 4557.8 ± 0.5 Ma are indistinguishable from one another and consistent with the less precise Sm-Nd ages (Lugmair and Galer, 1992); there is no indication that these ages might reflect anything other than the time of crystallization of these meteorites. Secondly, the range of the ^{55}Mn/^{52}Cr ratios is rather large, from ~0.5 in pyroxenes up to ~160 in LEW olivines, which affords the precise determination of the ^{53}Mn/^{55}Mn ratio at the time of isotopic closure.

We have measured the ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios in pyroxene (Px), olivine (Ol), and total rock (TR) samples from LEW and ADOR. The results are presented in Table 1 and Fig. 2. The listed ^{53}Cr excesses are recently refined values and are slightly different from the preliminary values published earlier (Lugmair et al., 1992). The ^{53}Cr excesses in Ol are corrected for a small spallation contribution using Cr production rates in the meteorite Grant (Birck, pers. commun.) and the exposure ages of LEW and ADOR of 17.6 Ma (Eugster et al., 1991) and 55.5 Ma (Lugmair and Marti, 1977), respectively.

Due to the high Mn/Cr ratio in LEW Ol the ^{53}Cr excess is large (~18 ε) while Px and TR with low Mn/Cr ratios exhibit only 0.5 to 0.7 ε. The data points form a well defined isochron (Fig. 2) whose slope corresponds to a ^{53}Mn/^{55}Mn ratio of $(1.25 \pm 0.07) \times 10^{-6}$ and an initial ^{53}Cr/^{52}Cr of 0.40 ± 0.16 ε at the time of isotopic closure. These values are reasonably close to those obtained by Nyquist et al. (1994): $(1.44 \pm 0.07) \times 10^{-6}$ and 0.40 ± 0.16 ε, respectively. The agreement becomes even better when the spallation contribution in the olivine sample of Nyquist et al. (1994) is taken into account and their ^{53}Mn/^{55}Mn is decreased to 1.40×10^{-6}. The Ol datum for ADOR is not precise enough to add any further constraint for the ^{53}Mn/^{55}Mn value. However, it is entirely consistent with the LEW data: the ADOR Ol data point falls well on the LEW best fit line (Fig. 2). The ADOR Px data are also concordant. The fact that the angrites cooled fast (Störzer and Pellas, 1977) suggests that both the U-Pb and the Mn-Cr isotope systems closed approximately at the same time, or, more correctly, within the time span afforded by the time resolution of our isotope chronometers. Thus, the obtained value of $(1.25 \pm 0.07) \times 10^{-6}$ represents the ^{53}Mn/^{55}Mn ratio at 4557.8 Ma ago. Using the angrites as absolute time markers we now can map the relative ^{53}Mn-^{53}Cr chronometer onto an absolute time scale.

3.1.2. Noncumulate eucrites

We continued our study of the ^{53}Mn-^{53}Cr isotope system with other early planetary differentiates, the basaltic achondrites (noncumulate eucrites). The reasons for choosing the eucrites were to establish that ^{53}Mn was still extant when these meteorites cooled from their parent magma and to precisely constrain the time of their formation. In addition, for the eucrite Chervony Kut (CK) we have previously obtained the first evidence for the presence in the early solar system of the extinct radionuclide ^{60}Fe ($T_{1/2}$ = 1.5 Ma; Shukolyukov and Lugmair, 1993a,b). Combining the information of an absolute age for CK (which is provided by the ^{53}Mn-^{53}Cr chronometer and the Pb-Pb age of the angrites as discussed above) with the relative abundance of ^{60}Fe in CK (and in other eucrites) enables us to estimate the initial ^{60}Fe/^{56}Fe in the early solar system. The ^{53}Mn-^{53}Cr systematics in some eucrites were originally published by Lugmair et al. (1994a,b) and Shukolyukov and Lugmair (1997a,b); some of the earlier results were refined recently and are updated here.

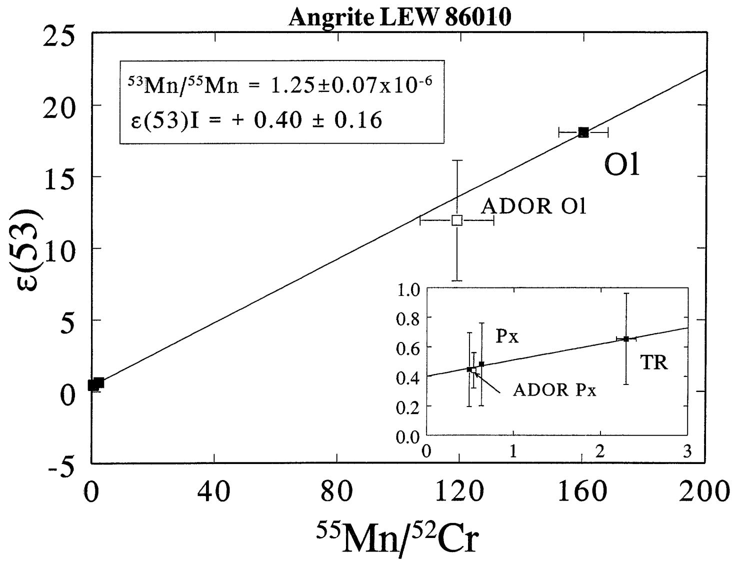

Fig. 2. ^{53}Mn-^{53}Cr systematics in the angrites LEW86010 (LEW; filled symbols) and Angra dos Reis (ADOR; open symbols). Here and in the following figures the ^{53}Cr/^{52}Cr excesses relative to the terrestrial value are expressed in ε-units (1 part in 10^4). TR - total rock, Ol- olivine, Px - pyroxene. The high Mn/Cr in LEW olivine ensures a good precision in the slope of the best fit line which corresponds to ^{53}Mn/^{55}Mn at the time of isotopic closure. The precise absolute Pb-Pb age of LEW (Lugmair and Galer, 1992) and the obtained ^{53}Mn/^{55}Mn ratio allow the relative ^{53}Mn-^{53}Cr chronometer to be mapped onto an absolute time scale. Although rather uncertain because of the very limited sample size, the ADOR olivine data point falls on the LEW best fit line which reflects a contemporaneous formation of the two angrites. The insert shows enlarged the results for the samples with small Mn/Cr ratios. The ADOR Px datum is totally consistent with the LEW data. ε(53)I is the initial ^{53}Cr/^{52}Cr at the time of isotopic closure.

Chervony Kut. We started our study of the eucrites with this moderately brecciated eucrite and measured the ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios in chromite (Chr), plagioclase (Pl), Px, and TR samples (Lugmair et al., 1994a). The most recent results are presented in Table 2 and in Fig. 3 . Although the ranges in the ^{55}Mn/^{52}Cr and ^{53}Cr/^{52}Cr ratios are very small because of the lack of olivine when compared with those in LEW (Fig. 2), the ^{53}Cr excesses correlate reasonably well with the respective ^{55}Mn/^{52}Cr ratios. This indicates that ^{53}Mn was still extant at the time of CK solidification. From the slope of the best fit line we derive the ^{53}Mn/^{55}Mn ratio of $(3.7 \pm 0.4) \times 10^{-6}$ and an initial ^{53}Cr/^{52}Cr of $0.49 \pm 0.09\ \varepsilon$ at the time of isotopic closure. The slope of the CK isochron is considerably steeper than that of LEW (shown for comparison by the dashed line in Fig. 3). If ^{53}Mn was homogeneously distributed in the region where the eucrite (EPB) and angrite parent bodies were formed then the difference in slopes corresponds to the time difference of 5.8 ± 0.8 Ma. Converting the relative ^{53}Mn-^{53}Cr age into an absolute age we obtain for CK 4563.6 ± 0.9 Ma. This age is surprisingly ancient. It is older than all precise absolute ages measured with modern techniques not only for eucrites but for any meteorite from any other class except Allende CAI's. Comparison of the CK age with that of the first solar condensates, Allende CAI's (4566 ± 2 Ma, Göpel et al., 1991), indicates that the processes of EPB accretion, core formation, mantle differentiation, and formation of the first basalts on the EPB occurred very early in the history of the solar system and took less than 5 Ma. The Pu-Xe and Pb-Pb ages for CK of 4538 ± 19 Ma (Shukolyukov and Begemann, 1996) and 4312.6 ± 1.6 Ma (Galer and Lugmair, 1996) are significantly younger than its ^{53}Mn-^{53}Cr age. This indicates diffusive losses of the radiogenic Xe and reequilibration of the U-Pb system during a secondary event(s).

Juvinas. We have studied another noncumulate eucrite containing vestiges of ^{60}Fe - Juvinas (JUV; Lugmair et al., 1994b). The newest data are given in Table 2 and Fig. 4. The range in the ^{55}Mn/^{52}Cr ratios in JUV samples is slightly smaller than that for CK. However, similar to CK, the measured ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios in Chr, Pl, Clast material (CL-A1) and TR samples from JUV correlate reasonably well (Fig. 4). The slope of the internal mineral isochron for JUV yields a ^{53}Mn/^{55}Mn ratio of $(3.0 \pm 0.5) \times 10^{-6}$ with an initial ^{53}Cr/^{52}Cr of $0.56 \pm 0.09\ \varepsilon$ at the time of its solidification. The slope of the JUV isochron is slightly smaller than that for CK which corresponds to a time difference of $1.1^{+0.9}_{-1.0}$ Ma of JUV relative to CK. This time difference appears to be smaller than that of $4.7^{+0.7}_{-1.0}$ Ma (or $4.7^{+1.1}_{-1.3}$ Ma, if the uncertainty in the half life of ^{60}Fe is taken into account) as indicated by the closure of the ^{60}Fe-^{60}Ni system in CK and JUV (Shukolyukov and Lugmair, 1993b). This may either indicate somewhat different closure temperatures or different sensitivities to secondary events for these two isotope systems and may be due to the high mobility of Ni (Shukolyukov and Lugmair, 1993b). Using the ^{53}Mn/^{55}Mn ratio of $(3.0 \pm 0.5) \times 10^{-6}$ in JUV and the Mn-Cr and Pb-Pb systematics of the angrites, the age of JUV relative to the angrites is 4.7 ± 0.9 Ma and the absolute age is then 4562.5 ± 1.0 Ma. This age is consistent with the less precise Pu-Xe age of 4551 ± 15 Ma (Shukolyukov and Begemann, 1996). The Pb-Pb age, however, is again much younger: 4320.9 ± 1.7 Ma (Galer and Lugmair, 1996) and indicates that the U-Pb system is sensitive to brecciation events and may have been totally reset during such an event.

Table 2. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in noncumulate eucrites

Sample*	Weight (mg)	Mn (ppm)	Cr (ppm)	^{55}Mn/^{52}Cr (±5%)	ε(53)
CK TR-1	216	3713	1706	2.46	1.29 ± 0.10
CK TR-5	70	3398	1645	2.34	1.29 ± 0.10
CK Chr	nd	–	–	0.024	0.51 ± 0.09
CK Px-2	203	7767	3972	2.21	1.18 ± 0.09
CK Pl	74	138	49.2	3.17	1.54 ± 0.10
JUV TR-2	232	4626	2188	2.38	1.22 ± 0.10
JUV TR-5	493	3844	1893	2.29	1.18 ± 0.10
JUV Chr	nd	–	–	0.011	0.57 ± 0.10
JUV Pl	40	98.5	38.3	2.90	1.33 ± 0.10
JUV Cl-A1	174	4657	1911	2.15	1.31 ± 0.18
IB TR	102†	4019	2726	1.66	0.95 ± 0.09
IB Chr	nd	–	–	0.015	0.79 ± 0.06
IB Sil	~102	4066	2546	1.80	0.97 ± 0.06
CAL TR	173	4326	2095	2.33	1.12 ± 0.09
CAL Chr	~0.3	9402	6.29×10^5	0.017	1.12 ± 0.09
CAL Px	55	7428	1072	7.82	1.15 ± 0.12
CAL Sil	161	4457	1000	5.03	1.11 ± 0.11
EET TR	50	4074	1171	3.93	1.12 ± 0.14
EET Chr	nd	–	–	0.019	1.16 ± 0.09
EET Aug	10	7886	2663	3.34	1.14 ± 0.10
EET Pig	60	7264	921	8.91	1.25 ± 0.09
POM TR	377†	4170	3237	1.45	0.71 ± 0.07
POM Chr	nd	–	–	0.02	0.73 ± 0.08
POM Sil	~377	4221	1795	2.65	0.74 ± 0.08

*Chervony Kut (CK), Juvinas (JUV), Ibitira (IB), Caldera (CAL), EET 87520 (EET), Pomozdino (POM), total rock (TR), chromite (Chr), pyroxene (Px), plagioclase (Pl), augite (Aug), pigeonite (Pig), clast (Cl), silicate portion (Sil).

†After fractional dissolution of a bulk sample equal proportions of the obtained solutions for "Sil" and "Chr" were recombined to produce a representative "TR" sample solution.

Ibitira. Ibitira (IB) is one of a few unbrecciated eucrites. Its vesicular texture suggests that this eucrite formed by rapid cooling either of a lava flow extruded onto the surface of the EPB (Wilkening and Anders, 1975) or of a pool of impact melt. However, its primary texture was blurred by strong static and shock metamorphism (Steele and Smith, 1976). The measured ^{53}Cr excesses in Chr, silicate (Sil) (which in eucrites consists mostly of a mixture of Px and Pl and is obtained by a dissolution of bulk samples in an HF/HNO_3 mixture leaving chromites behind), and TR samples and the respective ^{55}Mn/^{52}Cr

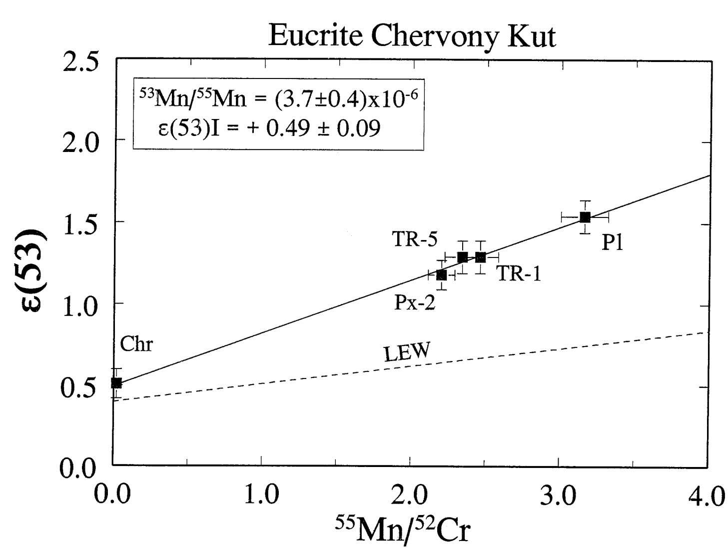

Fig. 3. ^{53}Mn-^{53}Cr systematics in the eucrite Chervony Kut (CK). The slope of the best fit line for the total rock samples (TR) and mineral fractions plagioclase (Pl), pyroxene (Px), and chromite (Chr) of CK is clearly larger than that for LEW (dashed line). The corresponding difference in ^{53}Mn/^{55}Mn yields a time difference of 5.8 ± 0.8 Ma. Converting the relative ^{53}Mn-^{53}Cr age into an absolute age (see text), the age of CK is 4563.6 ± 0.9 Ma. This age is older than all precise absolute ages measured so far for meteorites.

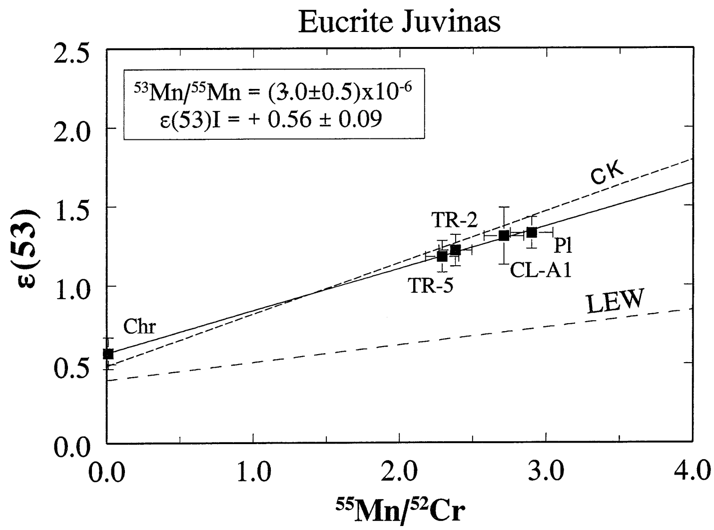

Fig. 4. ^{53}Mn-^{53}Cr systematics in the eucrite Juvinas (JUV). The slope of the best fit line for the total rock samples, clast (Cl-A1), and mineral fractions of JUV is slightly smaller than that for CK (short dashed line). The corresponding time difference is $1.1^{+0.9}_{-1.0}$ Ma. The data for LEW is shown for comparison (dashed line). The age of JUV relative to the angrites is 4.7 ± 0.9 Ma and the absolute age is then 4562.5 ± 1.0 Ma.

ratios are given in Table 2. The data points form an isochron (Fig. 5) whose slope yields a ^{53}Mn/^{55}Mn ratio of (1.06 ± 0.50) $\times 10^{-6}$ with an initial ^{53}Cr/^{52}Cr ratio of 0.79 ± 0.06 ε. The absolute age of IB, calculated analogous to that of CK and JUV, is 4557^{+2}_{-4} Ma. The time resolution is less than in the case of CK and JUV because the slope of the IB isochron is much flatter, and the uncertainties in ^{53}Cr/^{52}Cr become more significant. The age of IB is clearly younger than that of CK and JUV. If magmatism on the EPB occurred only at a very early stage, this age is consistent with a reequilibration of the ^{53}Mn-^{53}Cr system in IB. The obtained ^{53}Mn-^{53}Cr age is in good agreement with the Pb-Pb ages of 4556 ± 6 Ma (Chen and Wasserburg, 1985) and 4560 ± 3 Ma (Manhés et al., 1987). If indeed true, the very young ^{147}Sm-^{143}Nd age of 4460 ± 30 Ma (Prinzhofer et al., 1992) may be the result of post-crystallization secondary events although, generally, the Sm-Nd system is known to be more robust than the U-Pb system.

Caldera. Caldera (CAL) is the second unbrecciated eucrite in our set of samples. It differs from Ibitira in having a coarse grain size (typically a few mm) and no vesicles. The texture of CAL indicates slow cooling. The ^{53}Mn-^{53}Cr data for CAL (Table 2, Fig. 6) were published earlier by Wadhwa and Lugmair (1996). The analyzed samples, a chrome-spinel separate (Chr), Px, Sil, and TR, exhibit a wide range of ^{55}Mn/^{52}Cr

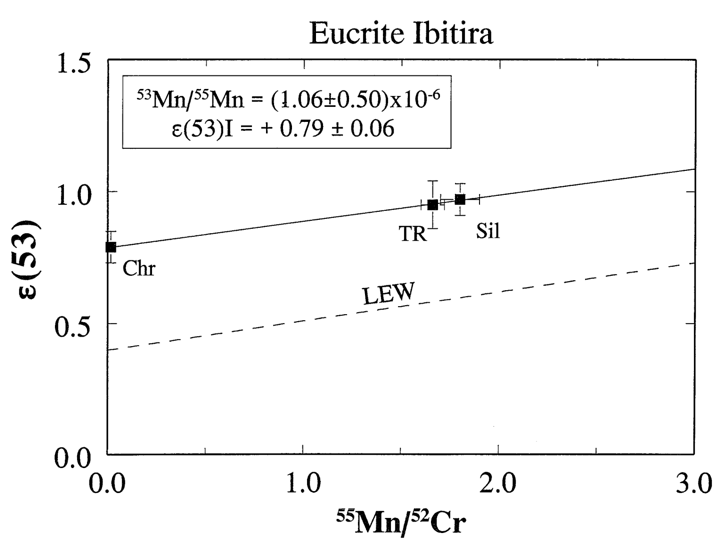

Fig. 5. ^{53}Mn-^{53}Cr systematics in the eucrite Ibitira (IB). The best fit line for the total rock sample, chromite, and silicates (Sil)—mostly a mixture of Px and Pl—is almost parallel to that for LEW (dashed line). This indicates that the ^{53}Mn-^{53}Cr system in these two meteorites closed essentially at the same time. In contrast to LEW, however, because of the lack of a phase with a high Mn/Cr ratio the absolute age of IB of 4557^{+2}_{-4} Ma is much less precise than that for LEW.

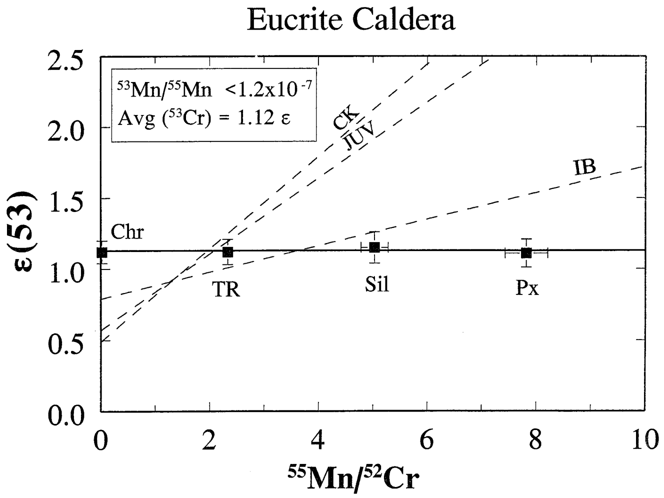

Fig. 6. ^{53}Mn-^{53}Cr systematics in the eucrite Caldera (CAL) (data from Wadhwa and Lugmair (1996)). In contrast to CK, JUV, and IB (dashed lines) the ^{53}Cr/^{52}Cr ratios in CAL samples are the same (the total range is only 4 ppm) in spite of a relatively wide range in Mn/Cr ratios: the best fit line has a zero slope. This indicates that CAL has equilibrated Chromium isotopes and that ^{53}Mn had practically completely decayed when the ^{53}Mn-^{53}Cr system closed in this meteorite. Calculated from the uncertainties the upper limit for ^{53}Mn/^{55}Mn corresponds to an age of ≤4545 Ma, that is, CAL is more than 13 Ma younger than LEW.

ratios, from ~0.015 in Chr to ~8 in Px. However, in contrast to CK, JUV, and IB the ^{53}Cr/^{52}Cr ratios in all samples are the same: the total range in ^{53}Cr/^{52}Cr is just 4 ppm with the average of 1.12 ε (Fig. 6). This means that CAL has equilibrated Chromium isotopes and indicates that ^{53}Mn had practically fully decayed by the time when the ^{53}Mn-^{53}Cr system closed in this meteorite. From the listed uncertainties an upper limit for ^{53}Mn/^{55}Mn of 1.2×10^{-7} can be calculated. This implies that the age of CAL is ≤4545 Ma, more than 19 Ma younger than CK. This young age may represent either a cooling age, or may reflect the process of impact melting that reequilibrated the Chromium isotopes in CAL or a combination of both. The Pu-Xe age of ~4513 Ma (Shukolyukov and Begemann, 1996) and the Pb-Pb age of 4516.1 ± 2.8 Ma (Galer and Lugmair, 1996) are consistent with the calculated upper limit of ≤4545 Ma. However, these ages are younger than that deduced from the combined ^{146,147}Sm-^{142,143}Nd and ^{53}Mn-^{53}Cr systematics in CAL (4537 ± 12 Ma; Wadhwa and Lugmair, 1996). This again indicates a significantly higher sensitivity of the Pu-Xe and U-Pb systems to secondary events.

EET87520. Mason (1988) has classified EET87520 (EET) as an unbrecciated eucrite similar in texture and mineral compositions to the cumulate eucrite Moore County. Nevertheless, in terms of chemistry, not texture, we have found several indications which favor a noncumulate origin of this eucrite. The concentrations of Sm, Nd, and Rb in bulk EET are 2.7 ppm, 8.9 ppm, and 0.24 ppm, respectively. This is totally consistent with the mean Sm, Nd, and Rb concentrations in noncumulate eucrites of 2.8 ppm, 8.9 ppm, and 0.30 ppm (Warren, 1992; Eucrite Database, pers. comm.) while those in cumulate eucrites are lower by factors of 3–10. In addition, $\mu_1 = {}^{238}U/{}^{204}Pb$ in EET is 222 (Lugmair et al., 1991) which is similar to that in noncumulate eucrites (~150) but clearly different from that in cumulate eucrites (~14; Tera et al., 1997). Therefore, in chemical terms, we tentatively associate EET with the noncumulate eucrite group although additional studies on this subject are required.

We have measured the ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios in Chr, pigeonite (Pig), augite (Aug), and TR samples from EET (Table 2, Fig. 7). The ^{53}Cr excesses in Chr, Aug, and TR are essentially the same: the total range in ^{53}Cr/^{52}Cr is just 3 ppm. The ^{53}Cr excess in Pig (with the highest ^{55}Mn/^{52}Cr of ~9) is somewhat larger but the uncertainties overlap. Although there is a hint for a nonzero slope in Fig. 7, from the available data only an upper limit for the ^{53}Mn/^{55}Mn ratio of 2.5×10^{-7} can be obtained. This limit corresponds to an age of ≤4549 Ma. The ^{147}Sm-^{143}Nd system is slightly disturbed in the minerals; omission of the Pig data point yields an age of 4547 ± 9 Ma (Lugmair et al., 1991). In contrast, the ^{146}Sm-^{142}Nd system is not affected by a disturbance and yields the ^{146}Sm/^{144}Sm ratio of 0.00694 ± 0.00045 (Lugmair et al., 1991). This value is the same as was measured in LEW, which has a Sm-Nd age of 4.55 Ga and a precise Pb-Pb age of 4557.8 Ma (Lugmair and Galer, 1992). Combining the data obtained from the ^{146,147}Sm-^{142,143}Nd and ^{53}Mn-^{53}Cr chronometers yields (following the procedure suggested by Wadhwa and Lugmair (1996)) the most likely age for EET of 4547–4549 Ma.

Pomozdino. Pomozdino (POM) is an anomalous, high-MgO/FeO, REE-rich, brecciated eucrite (Warren et al., 1990). The results obtained for Chr, Sil, and TR samples from POM are presented in Table 2 and Fig. 8. Similar to CAL and EET, the ^{53}Cr excesses in the POM samples are the same within the uncertainties although they are distinctly smaller than in the preceding eucrites. The upper limit for the ^{53}Mn/^{55}Mn ratio is 6×10^{-7} corresponding to an age of ≤4554 Ma. Thus, this meteorite cooled or reequilibrated its Chromium isotopes more than 10 Ma after the closure of the ^{53}Mn-^{53}Cr system in CK. The upper limit for the ^{53}Mn-^{53}Cr age of POM is consistent with its Pu-Xe age of 4559 ± 15 Ma (Shukolyukov and Begemann, 1996).

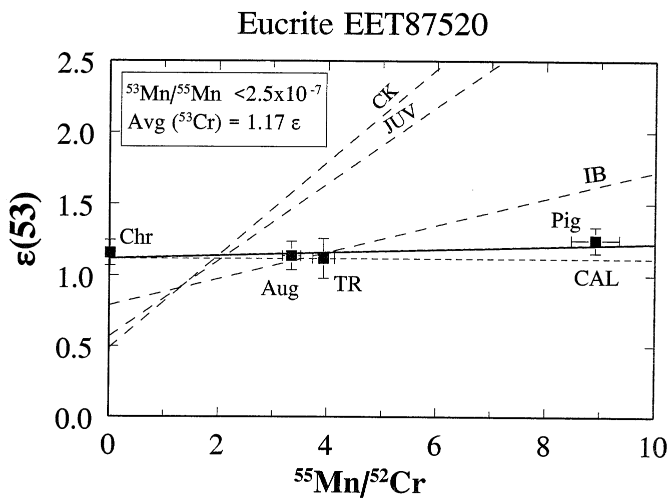

Fig. 7. ^{53}Mn-^{53}Cr systematics in the eucrite EET87520 (EET). The ^{53}Cr excesses in EET are very similar to those in Caldera and the slope of the best fit line for the total rock and mineral fractions (Aug - augite; Pig - pigeonite) is close to zero. Pig, with the highest Mn/Cr, reveals a slightly larger ^{53}Cr excess; however, the uncertainties overlap and only an upper limit for ^{53}Mn/^{55}Mn can be obtained. This limit corresponds to an age of ≤4549 Ma.

3.2. Heterogeneity of Radiogenic ^{53}Cr

While calculating the ^{53}Mn-^{53}Cr ages in the previous section we implied that the distribution of ^{53}Mn in the solar nebula was homogeneous, at least in the region where the angrite and eucrite parent bodies formed. However, as was emphasized in the introduction, the assumption of a homogeneous ^{53}Mn distribution upon which this chronology is based has to be tested. Our earlier results (Lugmair et al., 1992; Lugmair et al., 1994ab) have shown that the initial ^{53}Cr/^{52}Cr ratios in the angrites and the ancient eucrites CK and JUV are distinctly higher than the terrestrial value by 0.4–0.6 ε units. The bulk ε(^{53}Cr) values in CK and JUV are even twice as high as that in the angrite LEW but with a Mn/Cr ratio of about three times chondritic. Thus, this difference was originally deemed to be due to Mn/Cr fractionation in the parent bodies at the very early stages of their evolution when ^{53}Mn was still extant, possibly during the core formation process. However, a more quantitative reevaluation of the plausibility for this to occur (Drake et al., 1989; Drake, 1995, pers. commun.) has shown that only a small part of this difference in the ^{53}Cr/^{52}Cr ratios could possibly be attributed to a planetary differentiation. One possible explanation for the low terrestrial ^{53}Cr/^{52}Cr ratio is that the Earth is depleted in Mn relative to Cr. However, if this were the case, the Mn depletion must have occurred during the very early stages of the Earth's evolution or, more correctly, taking into account the short half life of ^{53}Mn, during the accretion

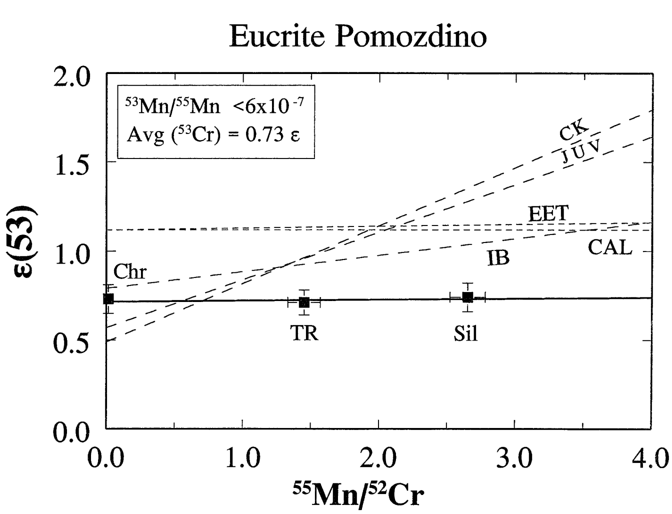

Fig. 8. ^{53}Mn-^{53}Cr systematics in the eucrite Pomozdino (POM). Similar to CAL and EET, the ^{53}Cr excesses in all POM samples are the same within the uncertainties (total range is 2 ppm). However, they plot below the CAL and EET isochrons which is probably due to a lower Mn/Cr ratio in bulk POM than in CAL and EET and their precursor material. The upper limit for ^{53}Mn/^{55}Mn corresponds to an age of ≤4554 Ma.

Table 3. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in chondrites

Sample*	Weight (mg)	Mn (ppm)	Cr (ppm)	^{55}Mn/^{52}Cr (±5%)	$\epsilon(53)$
DIM TR	1198†	2541	3794	0.76	0.53 ± 0.12
DIM Chr	nd	–	–	0.009	0.39 ± 0.09
DIM Sil	1153	2073	1353	1.73	0.59 ± 0.10
PLA TR	1159†	2648	3876	0.77	0.48 ± 0.15
PLA Chr	nd	–	–	0.011	0.39 ± 0.10
PLA Sil	980	2712	1250	2.45	0.65 ± 0.13
FIN TR	923†	2645	3881	0.77	0.52 ± 0.10
FIN Chr	nd	–	–	0.025	0.44 ± 0.10
FIN Sil	880	2454	1132	2.45	0.52 ± 0.10

*Dimmitt (DIM), Plainview (PLA), Finney (FIN), total rock (TR), chromite (Chr), silicate portion (Sil).
†After fractional dissolution of a bulk sample equal proportions of the obtained solutions for “Sil” and “Chr” were recombined to produce a representative “TR” sample solution.

and evolution of its precursor planetesimals. The other possible reason is that the Earth is actually anomalous and the relative lack of ^{53}Mn decay products on the Earth was caused by a heterogeneous radial distribution of ^{53}Mn which existed during the late stage of the solar nebula. In this regard it is important to establish the ^{53}Mn-^{53}Cr systematics of chondrites because chondrites, or chondrite-like material, are commonly believed to have formed the Earth, the other terrestrial planets and the parent bodies of the differentiated meteorites. If the elevated ^{53}Cr/^{52}Cr ratios in the angrites and eucrites are the result of an indigenous ^{53}Mn heterogeneity in the solar system, and if chondrites collected on the earth ultimately originate from the same region of the solar system as differentiated meteorites then chondrites should also carry a similar isotopic signature. One important difference between the chondrites and the differentiated meteorites is that there was no planet-wide Mn/Cr fractionation during the history of the chondrite parent planetesimals. Therefore, any variations of ^{53}Cr/^{52}Cr ratios in large bulk samples of chondrites cannot be attributed to changes of the Mn/Cr ratio in the parent bodies and should be characteristic for the whole parent bodies and for the region where these parent bodies formed. To this end we have studied the ^{53}Mn-^{53}Cr systematics in three ordinary chondrites. The original data were published by Lugmair and MacIsaac (1995).

3.2.1. Chondrites

We have analyzed the chondrites Dimmitt (H3,4), Plainview (H5), and Finney (L5). The results are presented in Table 3 and Fig. 9. The primary goal of the study of the chondrites was to obtain ^{53}Cr/^{52}Cr and Mn/Cr data on relatively large bulk sam-

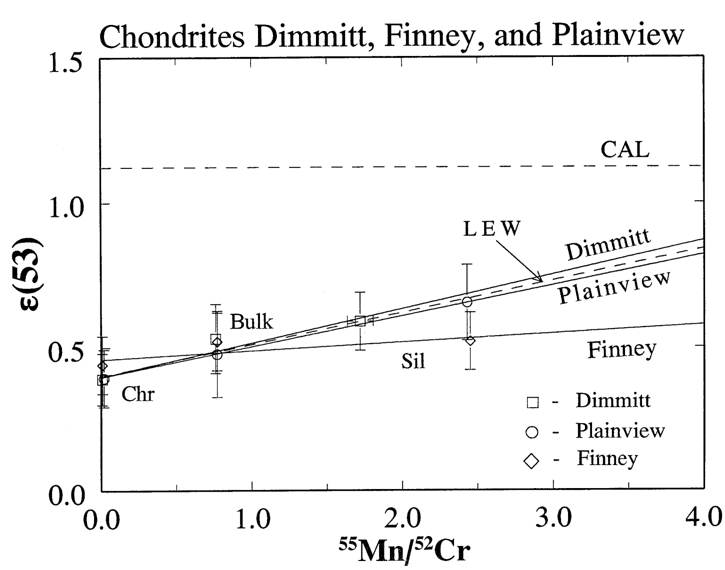

Fig. 9. ^{53}Mn-^{53}Cr systematics in the chondrites Dimmitt (DIM), Plainview (PLA), and Finney (FIN). The slopes of the best fit lines for DIM and PLA are indistinguishable from that for LEW (dashed line). However, because DIM and PLA are only partially equilibrated these data may not have chronological meaning (see text). FIN has an effectively equilibrated Chromium isotopic composition indicating isotopic closure at a time when ^{53}Mn had practically fully decayed. Bulk ^{53}Cr/^{52}Cr in chondrites of ~0.48 ε (with ^{55}Mn/^{52}Cr = 0.76) is clearly distinct from that in the Earth. Therefore, the main portion of the Earth must have accreted from a material which differs in ^{53}Mn/Cr from ordinary chondrites. The similarity of the ^{53}Mn-^{53}Cr systematics in chondrites and angrites indicates a close relationship between these classes of meteorites. Bulk chondrites, however, plot significantly below eucrites (the CAL best fit line is shown for comparison).

Table 4. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in shergottites and a lunar sample.

Sample*	Weight (mg)	Mn (ppm)	Cr (ppm)	^{55}Mn/^{52}Cr ($\pm$5%)	ϵ(53)
ALH Chr	nd	–	–	0.01	0.22 ± 0.10
ALH Px	65	3982	3176	1.42	0.23 ± 0.10
ALH Sil	60	3732	4891	0.86	0.21 ± 0.10
SHE TR	71	4482	1594	3.17	0.23 ± 0.07
EETA TR	49	3718	4078	1.03	0.27 ± 0.07
60025 Chr	nd	–	–	0.02	0.01 ± 0.09
60025 MAF	86	2909	1192	2.76	0.00 ± 0.10
60025 Ol	59	4025	56	81.1	-0.02 ± 0.15

*ALH 84001 (ALH), Shergotty (SHE), EETA 79001 (EETA), total rock (TR), chromite (Chr), pyroxene (Px), olivine (Ol), silicate portion (Sil), mafic fraction (MAF); 60025 is a lunar ferroan anorthosite.

ples and compare them with those in the differentiated meteorites and the Earth.

For Dimmitt and Plainview the ^{53}Cr excesses in chrome-spinel fractions (Chr) are slightly smaller than in the bulk samples while those for silicate portions (Sil) are slightly higher suggesting a correlation with their respective ^{55}Mn/^{52}Cr ratios. Interestingly, the slope of best fit lines and the initial ^{53}Cr/^{52}Cr ratios for both meteorites are indistinguishable within the uncertainties from those of LEW (shown in Fig. 9 by the dashed line). These chondrite data, however, may not have chronological meaning in a strict sense. Indeed, since Dimmitt and Plainview are only partially equilibrated, the best fit lines may simply reflect a mixture of different equilibration times for the individual chondritic components. Thus, to ascribe to these data real ages would be premature, which, in addition, was not the goal of these experiments. The best fit line for Finney has a slope which is indistinguishable from zero. This indicates that chromium isotopes were equilibrated at a time when ^{53}Mn had practically fully decayed.

The measured ^{53}Cr excesses in the bulk samples are clearly different from the terrestrial value by $\sim$0.5 ε. Within the listed uncertainties the values for all three meteorites are the same. Thus, ordinary chondrites possess uniform and higher than terrestrial bulk ^{53}Cr/^{52}Cr ratios. We, therefore, conclude that the precursor material of the Earth contained significantly less ^{53}Mn relative to Cr than did the chondrites (and the studied achondrites). The main portion of the Earth must have accreted from a material which is different from ordinary chondrites (at least in terms of the ^{53}Mn/Cr ratio). The ^{53}Mn-^{53}Cr systematics of chondrites is similar to that of the angrites which indicates a close relationship between these meteorites. Indeed, at the chondritic ^{55}Mn/^{52}Cr ratio of 0.76 (as defined from the measured values in Table 3) the corresponding ^{53}Cr excess for the angrites ($\sim$0.48 ε) is the same as the bulk chondrite value ($\sim$0.48 ε). This is consistent with a chondritic Mn/Cr ratio in the angrite parent body and with the same ^{53}Mn abundances in the angrite and the chondrite parent bodies and their precursors. However, there still remains an apparent difference between the chondrites and the eucrites which will be discussed below.

If the observed difference in the ^{53}Cr/^{52}Cr ratios between the Earth and the meteorites is indeed due to a radial gradient of ^{53}Mn distribution in the late stage solar nebula then samples which originate somewhere between the orbits of the Earth (1 AU) and the chondrites (asteroid belt, 2–3 AU) should show an intrinsic ^{53}Cr/^{52}Cr ratio which is intermediate between that of the chondrites and the Earth. Fortunately, this hypothesis can be tested: if the SNC meteorites indeed originate from Mars (at 1.51 AU) as is now widely assumed, then their ^{53}Cr/^{52}Cr ratios should be intermediate between that of the Earth (0 ε) and the chondrites and angrites ($\sim$0.48 ε). On the other hand, the low ^{53}Cr/^{52}Cr ratio on Earth could be due to the lower than chondritic Mn/Cr ratio in its silicate portion (Dreibus and Wänke, 1979, 1980). This low Mn/Cr ratio could have evolved in its precursor material due to the difference in volatilities between the two elements. In contrast, geochemical models for Mars based on SNC meteorites result in close to chondritic Mn/Cr ratios for this planet (Dreibus and Wänke, 1979, 1980; Drake et al., 1989; see Section 3.3 for further discussion). Thus, no ^{53}Mn/^{55}Mn gradient would be expected since SNC meteorites should then show chondritic ^{53}Cr/^{52}Cr excesses.

3.2.2. Mars and the Moon

We have studied the ^{53}Mn-^{53}Cr systematics in several SNC meteorites: samples from the pyroxenite ALH84001 (ALH), bulk Shergotty (SHE), and EETA79001, lithology A (EETA) (Table 4 and Fig. 10; Lugmair et al., 1996). Since ^{53}Mn was practically no longer extant by the time when formation of the terrestrial planets was completed (several tens to hundred of million years after the beginning of the solar system) for samples from Mars one could expect the same ^{53}Cr excesses regardless of their Mn/Cr ratios. Indeed, all mineral fractions from ALH, Chr, Px, and Sil, exhibit the same ^{53}Cr/^{52}Cr ratios ($\sim$0.22 ε) indicating total equilibration of Chromium isotopes after all ^{53}Mn had decayed. The same excess was also found in the bulk samples of SHE ($\sim$0.23ε). This value can be interpreted as the characteristic ^{53}Cr excess for Mars. Obviously, this excess of 0.22 ε is intermediate between that for the chondrites and the Earth (Fig. 10) and is consistent with the condition for a radial gradient of ^{53}Mn distribution in the late solar nebula as was discussed above. Furthermore, this characteristic ^{53}Cr excess provides an additional isotopic tool to distinguish SNC meteorites from other classes.

Note, that EETA exhibits a slightly higher ^{53}Cr excees ($\sim$0.27ε) than ALH and SHE. Although the present uncertainties overlap, the EETA results tend to be consistently higher in all measurements. If this difference should turn out to be true (the resolution of this question needs a great deal of further work) then the EETA source must have separated from the

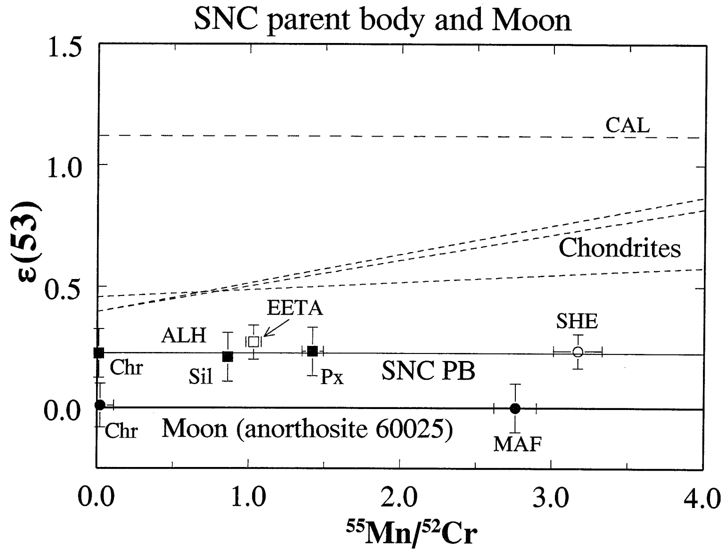

Fig. 10. ^{53}Mn-^{53}Cr systematics in the SNC parent body and in the Moon. Data for chondrites and eucrites (CAL) are shown for comparison. The mineral fractions from ALH84001 (ALH) and a bulk sample of Shergotty (SHE) reveal the same ^{53}Cr/^{52}Cr excess of ~0.22 indicating total equilibration of Chromium isotopes after all ^{53}Mn had decayed. This excess appears to be the characteristic ^{53}Cr excess for the SNC PB (Mars), is intermediate between that for the chondrites and the Earth, and satisfies the condition for a radial gradient of ^{53}Mn distribution in the late solar nebula. The sample of EETA79001, lithology A, (EETA) shows a slightly larger ^{53}Cr excess (~0.27 ε) , however, the uncertainties overlap (see text). ^{53}Cr excesses in the mafic portion (MAF) and mineral fractions from the lunar anorthosite 60025 (Ol with ^{55}Mn/^{52}Cr = 81 and with ε(53) ≈ −0.02, indistinguishable from the terrestrial Cr, plots far outside and to the right of the graph) indicate an identical ^{53}Cr/^{52}Cr isotopic signature for the Earth and the Moon. This finding represents new isotopic evidence for a close genetic relationship between the two planets.

Mars mantle earlier than those of ALH and SHE, which is also indicated by the ^{187}Hf - ^{187}W data (Lee and Halliday, 1997).

To determine the characteristic ^{53}Cr/^{52}Cr ratio in the Moon which, like the Earth, is located at 1 AU, we have studied the ancient lunar anorthosite 60025 and measured ^{53}Cr excesses in the mafic portion (MAF), Chr, Ol, and Pl (Table 4 and Fig. 10). The amount of Cr extracted from the Pl fraction turned out to be too small for obtaining the required precision and, therefore, Pl data are omitted here. The data for MAF, Chr, and Ol all show identical ^{53}Cr/^{52}Cr isotopic signatures as is to be expected from its Sm-Nd age (Carlson and Lugmair, 1988). More significantly, these ^{53}Cr/^{52}Cr ratios are the same as for the Earth. The data are also in agreement with the less precise results obtained for the lunar sample 15555 by Harper and Wiesman (1992). One important aspect of this finding lies in the fact that it represents new isotopic evidence (in addition to the oxygen data, Clayton and Mayeda, 1975) for a close genetic relationship between the two planets. It is consistent with the hypothesis that the Moon formed to a large part out of material from the Earth's mantle during a giant impact event (e.g., Cameron and Ward, 1976). Any large (~2/3) contribution of material from an impactor which originated at a distance from the sun much different from that of the Earth should be resolvable in ^{53}Cr/^{52}Cr.

3.2.3. *HED parent body*

The studies of samples from the Moon, Mars, and chondrites described above have shown the presence of a radial gradient in the abundances of radiogenic ^{53}Cr: the Earth and the Moon at 1 AU are characterized by the same ^{53}Cr/^{52}Cr ratios (0 ε), Mars at 1.51 AU exhibits a ^{53}Cr excess of ~0.22 ε and the ordinary chondrites from the asteroid belt (~2–3 AU) reveal a ^{53}Cr excess of ~0.48 ε. The eucrites indicate even larger ^{53}Cr excesses (Fig. 2–7). It is well known that the most likely parent asteroid for the eucrites is Vesta 4 which is located at 2.36 AU. In Lugmair et al. (1996), based on the results from the eucrite Caldera which has equilibrated chromium isotopes (section 3.1.2), we had assigned to the HED parent body (HED PB) a global ^{53}Cr excess of ~1.12 ε, consistent with the other eucrites CK and JUV. Using this simplified approach we plotted the average intrinsic ^{53}Cr excesses vs. the heliocentric distances of the sample sources where they are known (Earth, Mars, and Vesta). Placing the ^{53}Cr excess of ~0.5 ε for the chondrites onto a curve fitted through the data points yielded a radial distance for the location of the chondrite parent bodies at ~1.9 AU, the inner edge of the asteroid belt (Lugmair et al., 1996). However, as was noted by these authors, if Mn/Cr has fractionated very early in the history of the HED PB, with a bulk model Mn/Cr ratio close to chondritic, as suggested by Dreibus and Wänke (1979, 1980) and Drake et al. (1989), this planetesimal may actually have a ^{53}Cr excess much smaller than 1.12 ε. (But note that if the apparent difference in the characteristic ^{53}Cr excesses between the eucrites and the angrites and chondrites were real the ages calculated in section 3.1.2 would be invalid because of the difference in the original ^{53}Mn abundances). To learn more about the evolution of the HED PB and to constrain further the extent of possible ^{53}Mn heterogeneity within the asteroid belt itself and, thus, to determine whether

Table 5. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in diogenites and cumulate eucrites

Sample*	Weight (mg)	Mn (ppm)	Cr (ppm)	^{55}Mn/^{52}Cr ($\pm$5%)	ϵ (53)
JT TR	422†	3868	5974	0.73	0.58 $\pm$ 0.09
JT Chr	nd	–	–	0.003	0.58 $\pm$ 0.05
JT Sil	~422	3721	5136	0.82	0.62 $\pm$ 0.08
SHA TR	173†	4526	12738	0.40	0.39 $\pm$ 0.12
SHA Chr-1	nd	–	–	0.011	0.38 $\pm$ 0.09
SHA Chr-2	nd	–	–	0.009	0.39 $\pm$ 0.09
SHA Sil	~173	4489	3636	1.39	0.39 $\pm$ 0.09
MC Chr	nd	–	–	0.020	0.89 $\pm$ 0.07
MC Sil	213	3961	1533	2.92	0.86 $\pm$ 0.07
SM Chr-1	nd	–	–	0.016	0.53 $\pm$ 0.08
SM Chr-2	nd	–	–	0.015	0.59 $\pm$ 0.09
SM Px	40	6614	1644	4.54	0.77 $\pm$ 0.07

*Johnstown (JT), Shalka (SHA), Moore County (MC), Serra de Magé (SM), total rock (TR), chromite (Chr), pyroxene (Px), silicate portion (Sil); for both SHA and SM the Chr-1 fractions are coarse grained while Chr-2 is fine grained.

†After fractional dissolution of a bulk sample equal proportions of the obtained solutions for "Sil" and "Chr" were recombined to produce a representative "TR" sample solution.

the ^{53}Mn-^{53}Cr isotope system can be used as a chronometer for different asteroid belt materials, we have studied other constituents of the HED PB, diogenites and cumulate eucrites. The preliminary results were published by Lugmair and Shukolyukov (1997) and Shukolyukov and Lugmair (1997).

Diogenites. We have measured ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios in Chr, Sil, and TR samples from the diogenites Johnstown (JT) and Shalka (SHA) (Table 5 and Fig. 11). For SHA two Chr phases were analyzed: a fine grained Chr 1 and a coarse grained Chr 2. Since diogenites are thought to have formed relatively deep in the HED PB interior and, therefore, cooled slowly, one could expect an equilibrated chromium isotopic signature in the minerals. Indeed, both meteorites show equilibrated chromium isotopic composition—within the uncertainties the phases with different Mn/Cr ratios exhibit the same ^{53}Cr excesses. However, the measured ^{53}Cr excesses in JT (~0.59 ε) and SHA (~0.39 ε) are considerably lower than those in Caldera and EET. This indicates that Mn/Cr fractionation in the HED PB has occurred very early in the history of the solar system and that the ^{53}Cr excess in CAL (and EET) does obviously not represent that of the bulk HED PB. The sources of SHA and (probably) JT, which possess relatively low Mn/Cr ratios, must have separated at a time when a considerable part of ^{53}Mn was still extant. The upper limits for the ^{53}Mn/^{55}Mn ratio at the time of the isotope closure in JT and SHA are 2×10^{-6} and 1×10^{-6}, respectively.

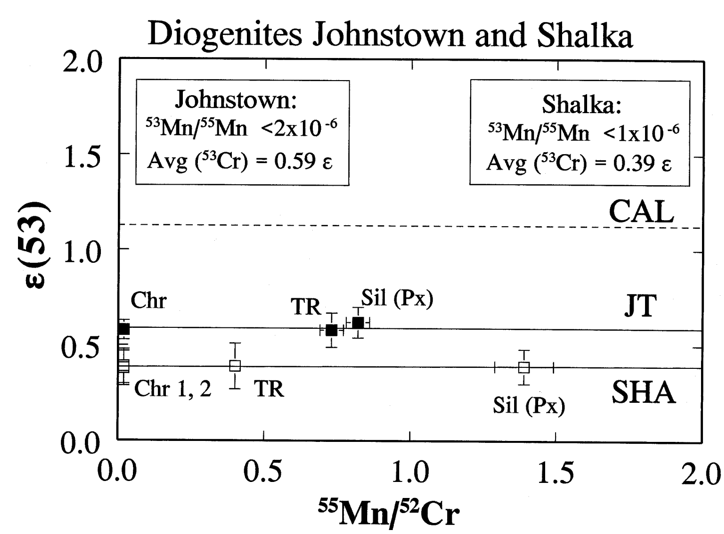

Fig. 11. ^{53}Mn-^{53}Cr systematics in the diogenites Johnstown (JT) and Shalka (SHA). Both meteorites exhibit an equilibrated Chromium isotopic signature. The ^{53}Cr excesses in JT (~0.59 ε) and SHA (~0.39 ε) are significantly lower than those in eucrites (CAL is shown for comparison). This indicates that Mn/Cr fractionation in the HED PB has occurred very early in the history of the solar system when a considerable part of ^{53}Mn was still extant. Since the total range of Mn/Cr in the diogenite samples is rather small, the upper limits for ^{53}Mn/^{55}Mn are very uncertain.

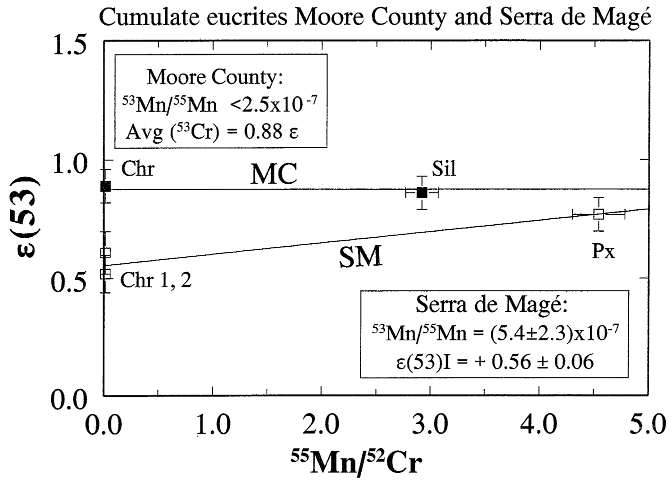

Fig. 12. ^{53}Mn-^{53}Cr systematics in the cumulate eucrites Moore County (MC) and Serra de Magé (SM). The ^{53}Cr excesses in the Chr and Sil fractions of MC are essentially the same and indicate that MC has equilibrated Chromium isotopes and ^{53}Mn was extinct when the ^{53}Mn-^{53}Cr system closed in MC. The upper limit for ^{53}Mn/^{55}Mn in MC corresponds to an age of ≤4549 Ma. In contrast to MC, the best fit line for the SM samples has a non zero slope which corresponds to an age of 4553^{+2}_{-4} Ma. This points towards comparable formation times for some cumulate and noncumulate eucrites.

Cumulate eucrites. We have studied two cumulate eucrites: Moore County (MC) and Serra de Magé (SM; Shukolyukov and Lugmair, 1997). The measured ^{53}Cr/^{52}Cr and ^{55}Mn/^{52}Cr ratios are given in Table 5 and Fig. 12 .

The ^{53}Cr excesses measured in Chr and Sil samples from MC are intermediate between those in most noncumulate eucrites and in the diogenites and are essentially the same: ~0.89 ε and ~0.86 ε, respectively. This indicates that MC has equilibrated chromium isotopes and ^{53}Mn was basically extinct when the ^{53}Mn-^{53}Cr system closed in MC. This is consistent with a young Sm-Nd age of 4456 ± 25 Ma (Tera et al., 1997). The upper limit for the ^{53}Mn/^{55}Mn is ~2.5×10^{-7} which corresponds to ≤4549 Ma.

In contrast to MC, the ^{53}Cr excesses in SM phases are different. We have analyzed two Chr and Px. The measured concentration of Cr in the Pl fraction turned out to be too small to allow a precise chromium isotope analysis. Chr 1 is coarse grained (mm size). Chr 2 is fine grained and is associated with Px. We found only a marginal difference in the Chromium isotopic compositions of the two chromites: ^{53}Cr excesses are ~0.52 ε and ~0.61 ε, respectively, while that in Px is ~0.77 ε. With ^{55}Mn/^{52}Cr ratios of 0.015 in Chr 1 and Chr 2, and of 4.54 in Px, the slope of the isochron yields a ^{53}Mn/^{55}Mn ratio of $(5.4 \pm 2.3) \times 10^{-7}$ and a corresponding age of 4553^{+2}_{-4} Ma. This suggests that the Mn-Cr system in SM closed relatively early in EPB history and that its young ^{147}Sm-^{143}Nd age of 4410 ± 20 (Lugmair et al., 1977) dates a secondary event. However, the ^{146}Sm/^{144}Sm ratio of 0.0068 ± 0.0016 in SM (Lugmair, 1977, unpubl. data) still reflects and is consistent with the older ^{53}Mn-^{53}Cr age of ~ 4553 Ma. (Note that the suggested value of the ^{146}Sm/^{144}Sm ratio at 4.56 Ma is 0.0070 ± 0.0010; Lugmair and Galer, 1992). The disparity of the ^{147}Sm-^{143}Nd and ^{146}Sm-^{142}Nd systematics in SM can possibly be explained using the model suggested by Prinzhofer et al. (1992). It has to be noted that there is a discrepancy in the ^{147}Sm-^{143}Nd ages of cumulate eucrites. The young ^{147}Sm-^{143}Nd ages cited above and that of Moama (4460 ± 30 Ma; Jacobsen and Wasserburg, 1984) suggest that cumulate eucrites formed ~100 Ma after noncumulate eucrites. However, these Sm-Nd ages disagree with those for MC and Moama of 4600±40 Ma (Nakamura et al., 1977) and 4580 ± 50 Ma (Hamet et al., 1978), respectively. Tera et al. (1997) has shown recently that the Pb-Pb ages for three cumulate eucrites are young and fall into the range of 4410–4480 Ma. Based on the cooling rates of cumulate eucrites of Miyamoto and Takeda (1994b), these authors concluded that young Pb-Pb ages of cumulate eucrites are not cooling ages but most likely represent their true crystallization ages. However, the former presence in SM of live ^{53}Mn strongly favors comparable formation times for cumulate and noncumulate eucrites. A high susceptibility of Pb-Pb ages to secondary events (as demonstrated for the noncumulate eucrites; section 3.1.2) may result in apparently young ages of cumulate eucrites which may also be fortuitously supported because of possible uncertainties in the cooling rates.

Bulk meteorite isochron for the HED PB. Figure 13 summarizes the results obtained for the bulk samples of all constituents of the HED PB studied so far. We plotted the measured ^{53}Cr/^{52}Cr ratios for the bulk meteorites vs. their respective ^{55}Mn/^{52}Cr ratios. In cases where only the data on mineral fractions were available (MC and SM) the ^{53}Cr/^{52}Cr ratios for the bulk samples were calculated from mass balance. This procedure was tested with the chondrites and it turned out that the measured and the calculated ^{53}Cr/^{52}Cr ratios are in very good agreement. To minimize the influence of sampling problems and to ensure that, for the samples with equilibrated chromium isotopic signature, the used bulk ^{55}Mn/^{52}Cr ratios are fairly representative, we have averaged the ^{55}Mn/^{52}Cr ratios obtained in this work and those from the literature. All data points form a well defined correlation line. The POM data point falls slightly below the correlation line but it still falls on the line within the uncertainty limits and is included in the calculation of its slope. If recommended modal Mn and Cr concentrations (Warren et al., 1990) are used, the POM data point falls even closer to the best fit line. The SM data point is shown in

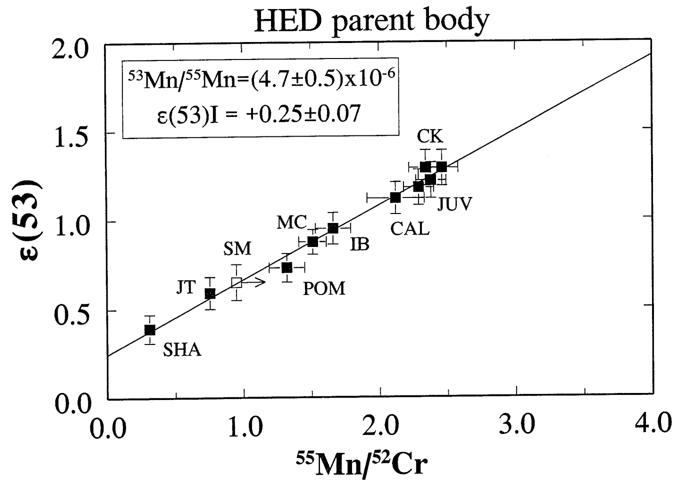

Fig. 13. ^{53}Mn-^{53}Cr systematics in the HED parent body. The measured $^{53}Cr/^{52}Cr$ ratios for the bulk meteorites are plotted vs. their respective $^{55}Mn/^{52}Cr$ ratios. For the samples with an equilibrated Chromium isotopic signature the average $^{55}Mn/^{52}Cr$ ratios of the values obtained in this work (tw) and those from the literature were used: CAL - (tw; Spettel, 1995, pers. commun.), JT - (tw; Floran et al., 1981; Fredriksson et al., 1976; Fukuoka et al., 1977; Mason and Jarosewich, 1971; McCarthy et al., 1972a), MC - (tw; McCarthy et al., 1972b; Spettel, 1995, pers. commun.), POM - (tw; Warren et al., 1990), SHA - (tw; Fukuoka et al., 1977; McCarthy et al., 1972a; Spettel, 1996, pers. commun.), SM - (Ma and Schmitt, 1979; Palme et al., 1978). The SM data point was not included in the calculation of the best fit line (see text). All data points form a well defined isochron whose slope dates the last Mn/Cr fractionation in the HED mantle - 4564.8±0.9 Ma ago. The absence of a resolvable scatter of the data points from the line implies contemporaneous formation of the sources of all these meteorites. The fact that the isochron passes close to the chondritic point (~0.48 at $^{55}Mn/^{52}Cr = 0.76$) suggests that the Mn/Cr ratio of the HED PB is close to chondritic. Since the bulk HED PB $^{53}Cr/^{52}Cr$ ($\varepsilon(53)$ at 0.76) of ~0.57 ε is only marginally higher than that in chondrites and in the angrites (~0.48 ε), there is no detectable heterogeneity in ^{53}Mn between the eucrite, angrite, and chondrite parent bodies. Thus, at least for samples from these parent bodies, the ^{53}Mn-^{53}Cr isotope system can be used as a chronometer.

Fig. 13 but was not included in the calculation. Since SM is a very coarse grained meteorite with a notoriously heterogeneous mineral composition the Mn/Cr ratio for SM is poorly constrained. The SM data point falls on the best fit line if the $^{55}Mn/^{52}Cr$ ratios of 0.97 (Palme et al., 1978) and 0.92 (Ma and Schmitt, 1979) are used. Other analyses (e.g., Jarosewich, 1990) indicate higher Mn/Cr ratios. (For a similar reason the bulk EET data were omitted from consideration. The weight of the analyzed bulk sample of this coarse grained eucrite was just 50 mg, and its high $^{55}Mn/^{52}Cr$ ratio (Table 2) is most likely not representative). We note that the SM data point on the bulk HED isochron falls between the diogenites and eucrites. This is consistent with the position of this meteorite on the multi-element mixing diagram of Dreibus and Wänke (1980): SM plots very close to howardites; this indicates close similarity of the chemical compositions and possibly suggests a genetic relationship between SM and howardites. The fact that SM has a relatively old ^{53}Mn-^{53}Cr age (~4553 Ma) indicates that it cooled fast at shallow depth. Taken together, these findings are consistent with the formation of this meteorite from a howardite type regolith during an impact event and, thus, SM may represent a secondary cumulate.

Because the best fit line is a bulk meteorite isochron it carries no information on the time of crystallization or cooling of individual meteorites. Instead, the slope of the line dates the last Mn/Cr fractionation and Chromium isotope equilibration in the HED mantle and, thus, the most likely time of core formation, and yields the $^{53}Mn/^{55}Mn$ ratio of $(4.7 \pm 0.5) \times 10^{-6}$ with an initial $^{53}Cr/^{52}Cr$ ratio of ~0.25 ε at this time. The absence of a resolvable scatter of the data points from the line implies that the source reservoirs of all these meteorites were formed contemporaneously and that the Mn-Cr systems of the bulk samples of these meteorites remained closed since their formation. We note, however, that the closer the $^{55}Mn/^{52}Cr$ ratio of a bulk meteorite falls to the chondritic value (0.76) the smaller becomes the time resolution. For example, material with a $^{55}Mn/^{52}Cr$ ratio of 0.8, which was separated from a chondritic source (see below) 3 Ma after the other meteorite sources were formed, would result in an offset from the isochron of only -0.01 ε and, thus, would go undetected. For this reason we cannot, for example, actually date the formation of the JT source: its Mn/Cr ratio of 0.77 is too close to the chondritic value. In contrast, the corresponding offset for a $^{55}Mn/^{52}Cr$ ratio of 2 would yield a negative offset of -0.25 ε which is quite measurable. In fact, Lugmair and Shukolyukov (1997) tentatively attributed the apparent positive offset of the CK data points to a slightly earlier separation of its source from the mantle. However, the $^{53}Cr/^{52}Cr$ data for CK were obtained several years ago and were not, until very recently, reevaluated for slight changes in machine bias as was done for LEW and ADOR. In contrast, the newly obtained data indicate that there is no detectable deviation of the CK data points from the best fit line.

We note (Fig. 13), that the isochron passes close to the chondritic point (~0.48 ε at 0.76). This indicates that the Mn/Cr ratio of the HED PB is indeed close to chondritic. The bulk HED PB $^{53}Cr/^{52}Cr$ ratio is ~0.57 ε and is only marginally higher than that in chondrites and that calculated for the ang-

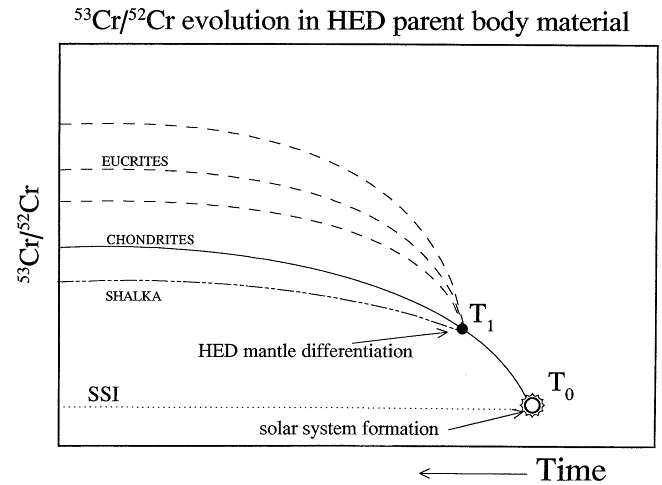

Fig. 14. The ^{53}Mn-^{53}Cr isotopic evolution in the HED parent body. T_0 is the time of addition to the solar nebula of live ^{53}Mn. SSI is the solar system initial ^{53}Cr/^{52}Cr ratio. Due to the decay of ^{53}Mn, the ^{53}Cr/^{52}Cr ratio in the HED PB precursors, posessing a chondritic Mn/Cr ratio, initially evolved along the chondrites curve. T_1 is the time of global fractionation of the HED PB mantle and formation of the source reservoirs of eucrites and diogenites. Since that time the source reservoirs and the bulk samples of the meteorites remained closed systems and their ^{53}Cr/^{52}Cr evolved depending on their respective Mn/Cr ratios.

rites (~0.48ε). Therefore, we conclude that the apparent difference in ^{53}Cr/^{52}Cr between the eucrites alone and the other non-SNC meteorites (see above) was misleading and that the higher ^{53}Cr/^{52}Cr ratios in eucrites are due to a very early Mn/Cr fractionation in the HED PB. The ^{53}Mn-^{53}Cr isotope system, therefore, can be used as a chronometer at least for samples from the eucrite, angrite, and chondrite parent bodies. We, thus, can further conclude that the absolute ^{53}Mn-^{53}Cr ages for eucrites (section 3.1.2) are valid. From the ^{53}Mn/^{55}Mn of $(4.7 \pm 0.5) \times 10^{-6}$ for the HED PB and that of LEW we calculate a relative time for the HED PB mantle fractionation of 7.1 ± 0.8 Ma before LEW. Using the absolute age of LEW we convert the relative age into an absolute age and obtain 4564.8 ± 0.9 Ma.

The evolution of ^{53}Cr/^{52}Cr in the HED PB is schematically illustrated in Fig. 14. T_0 corresponds to the onset of ^{53}Mn decay in the nebula. The curve marked chondrites describes the evolution in material with chondritic Mn/Cr ratio, in particular, in the HED PB precursors and in the bulk HED PB. At time T_1 a global fractionation of the HED PB mantle occurred and the sources of all HED meteorites were formed. The evolution of ^{53}Cr/^{52}Cr in eucrites is characterized by a series of curves whose steepness is proportional to their respective Mn/Cr ratios. Since eucrites have larger than chondritic Mn/Cr ratios the curves are located above the chondritic curve while the evolution curve for the diogenite SHA passes below. The JT curve is indistinguishable from the chondritic evolution path.

The study of the Mn-Cr systematics in two basaltic clasts (Pebble 16 and clast 4679) from the mesosiderite Vaca Muerta has shown that the data points for the bulk samples coincide with the HED isochron (Wadhwa et al., 1997b). This finding suggests that either these clasts originate from the HED parent body or that the mantles of the mesosiderite and the HED parent bodies, two similar but different planetesimals, differentiated at the same time, 4564.8 ± 0.9 Ma ago, and produced basaltic magmas. The second possibility appears to be less likely. In the contrary, it may imply that mesosiderites and the HED meteorites even come from the same parent body. Although at present, no definitive conclusion can be drawn with the limited set of data (Wadhwa et al., 1997b).

3.2.4. *Pallasites and Acapulco*

To extend the ^{53}Mn-^{53}Cr systematics to other meteorite classes and, further, to test the hypothesis that most meteorites originating from the asteroid belt had similar intrinsic ^{53}Mn/^{55}Mn ratios, we have studied two pallasites, Omolon (OM) and Springwater (SPR), and the primitive achondrite Acapulco (ACA) (see Zipfel et al., 1996).

Table 6. Manganese and chromium concentrations, Mn/Cr ratios, and ^{53}Cr excesses in pallasites

Sample*	Weight (mg)	Mn (ppm)	Cr (ppm)	^{55}Mn/^{52}Cr ($\pm$5%)	ϵ(53)
OM Chr	nd	–	–	0.011	1.28 ± 0.09
OM Ol-1	108	2268	384	6.67	1.21 ± 0.10
OM Ol-2	32	2337	464	5.69	1.07 ± 0.07
SPR Ol	64	1938	220	9.95	1.28 ± 0.09

*Omolon (OM), Springwater (SPR),chromite (Chr), olivine (Ol).

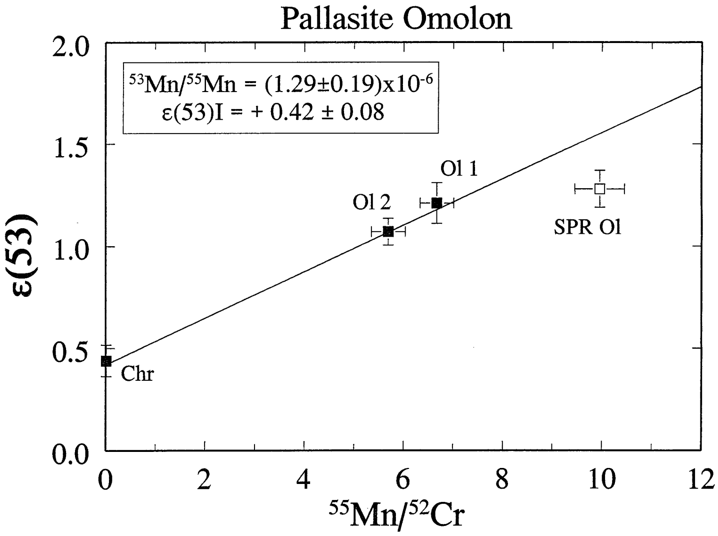

Fig. 15. ^{53}Mn-^{53}Cr systematics in the pallasites Omolon (OM; filled symbols) and Springwater (SPR; open symbol). The OM data points form an isochron whose slope yields a ^{53}Mn/^{55}Mn ratio indistinguishable from that in LEW. The corresponding age is 4558.0 ± 1.0 Ma. At the chondritic ^{55}Mn/^{52}Cr ratio of 0.76 the corresponding ^{53}Cr excess of ~0.52 ε is, within uncertainties, the same as that in chondrites (~0.48 ε). This suggests that the pallasite source material had most likely a chondritic Mn/Cr ratio and that there is no detectable heterogeneity of ^{53}Mn between the pallasites and the other meteorite classes from the asteroid belt studied so far. Assuming an initial ^{53}Cr/^{52}Cr for SPR of ~0.4 ε (see text), its absolute age is only slightly younger than that of OM and is ~4557 Ma.

Omolon. We have measured ^{53}Cr/^{52}Cr and Mn and Cr concentrations in Chr and in two olivine separates (Olv 1, Olv 2) (Shukolyukov and Lugmair, 1997a, Table 6 and Fig. 15). It has been shown by Zhou and Steele (1993) that Mn and particularly Cr in olivines from pallasites can be severely zoned. While the Mn content may slightly increase towards grain boundaries, the Cr content is highly depleted in the outer ~250 μm. This diffusive gain of Mn and loss of Cr by olivine may have occured sometime after the olivine had crystallized. Thus, measured Mn/Cr ratios and radiogenic ^{53}Cr excesses may be decoupled. In order to minimize this problem one either has to remove the outer ~250 μm from olivine grains before dissolution or use grains of sufficient size so that the diffusion effects in the surface layer are negligible. We have chosen the latter option and have selected for analysis large olivine grains of several mm in size. All three OM samples show excesses of ^{53}Cr which are correlated with the respective ^{55}Mn/^{52}Cr ratios. The ^{53}Cr excesses in the olivines are corrected for a small spallation contribution. Since the exposure age of OM is poorly constrained we have calculated the spallation contribution from the direct measurement of the Chromium isotopic composition in the adjacent metal phase (section 2.3). The data points form an isochron whose slope yields a ^{53}Mn/^{55}Mn ratio of (1.29 ± 0.19) × 10^{-6} at the time when the Mn-Cr system closed in OM. We note (Fig. 15), that similar to the angrites at the chondritic ^{55}Mn/^{52}Cr ratio of 0.76 the corresponding ^{53}Cr excess of ~0.52 ε is within uncertainties the same as that in chondrites (~0.48 ε). If this agreement is not coincidental, this suggests that the pallasite source material had a chondritic Mn/Cr ratio and that there is no detectable heterogeneity of ^{53}Mn between the pallasites and the other meteorite classes. Thus, the ^{53}Mn-^{53}Cr chronometer can be used for determination of an absolute time for the isotopic closure of the Mn-Cr system in OM. The ^{53}Mn/^{55}Mn of (1.29 ± 0.19) × 10^{-6} in OM is essentially the same as that in LEW of (1.25 ± 0.07) × 10^{-6}. OM is thus 0.2 ± 0.9 Ma older than LEW. Converting the relative age of OM into an absolute age we obtain 4558.0 ± 1.0 Ma.

The isotopic closure of the Mn-Cr system in OM at 4558 Ma implies a relatively fast cooling rate for pallasites (~50−100°C/Ma) at least in the high-temperature range. This is consistent with the results for pallasites of Miyamoto and Takeda (1994a) who, based on chemical zoning in olivine, deduced an extremely fast cooling rate in the range of 1100−600°C and a very shallow burial depth.

It was recently shown by Shen et al. (1997) that the ^{187}Re-^{187}Os isotope data for pallasites are consistent with those for IIAB irons: the data points for the pallasites fall on the well defined IIAB whole rock isochron which has a slope of 0.07848 ± 0.00018. Using the OM age of 4558 Ma, the Re-Os isotope data on pallasites, and assuming that the formation of OM was contemporaneous with the other pallasites and that the Re-Os system in the metal closed at the same time as Mn-Cr in Ol and Chr we calculate a decay constant $\lambda(^{187}\text{Re}) = 1.658 \times 10^{-11}$ a^{-1} (Shukolyukov and Lugmair, 1997a). However, there is still a dilemma when the steeper slopes for other classes of iron meteorites are considered (Shen et al., 1996). Using the obtained decay constant of 1.658 × 10^{-11} a^{-1} for the IVA-IVB group yields unreasonably old ages of close to 4.6 Ga. If the pronounced difference in age between this latter group of iron meteorites and the IIABs is really true then this would argue for an even higher decay constant of ^{187}Re. It would then also imply that the Re-Os system in the pallasites closed considerably later than the Mn-Cr system. These possibilities cannot be excluded at present. But because of the narrow range in Re/Os of the IVA-IVB data this difference in age is highly uncertain (Shen et al., 1996).

Springwater. The results of the analysis of an Ol sample are given in Table 6 and in Fig. 15. Since the exposure age of SPR is rather high (~100 Ma, Megrue, 1968) the excess of ^{53}Cr was corrected for a spallation contribution in a manner similar to OM. The lack of a chromite phase in SPR precludes the precise determination of its age. However, assuming that the initial ^{53}Cr/^{52}Cr ratio is similar to that of OM (~0.42 ε) one can calculate a ^{53}Mn/^{55}Mn ratio of ~9.8×10^{-7} which corresponds to a relative age of –1.3 Ma (i.e., 1.3 Ma after LEW) and to an absolute age of ~4556.5 Ma. Thus, SPR may be ~1.3 Ma younger than OM. Although we assumed the initial ^{53}Cr/^{52}Cr ratio for SPR, the choice of an initial is not crucial for calculating the age. Indeed, since the pallasite source had most likely chondritic Mn/Cr, the initial ^{53}Cr/^{52}Cr ratio cannot exceed the present day bulk chondritic ^{53}Cr/^{52}Cr ratio of ~ 0.48 ε. On the other hand, the fact that the SPR Ol data point falls below the OM isochron (Fig. 15) precludes the SPR initial ^{53}Cr/^{52}Cr to be lower than that of OM. One can calculate that using the total range of possible SPR initial values of 0.34 ε (the lower limit for the OM initial ^{53}Cr/^{52}Cr) up to ~0.52 ε (the measured ^{53}Cr excess in OM at chondritic Mn/Cr) results in a time difference of only ~1 Ma.

It has to be noted that the ^{53}Mn/^{55}Mn ratios in both SPR and OM are considerably smaller than those obtained on an ion microprobe with much less precision by Hutcheon and Olsen (1991) and by Hsu et al. (1997). The reason for this discrepancy is not clear at present. Aside from the possibility of unknown systematic errors which was noted by Hsu et al. (1997), ion microprobe analyses may sample areas of local heterogeneities of the Mn-Cr distribution (i.e., zones of very high Mn/Cr ratios) which in TIMS measurements, on a more macroscopic scale, are averaged out.

The ^{53}Mn/^{55}Mn ratios in OM and SPR are a factor of 2–3 smaller than in the pallasite Eagle Station (ES) (~2.3×10^{-6}) as measured by Birck and Allègre (1988). The corresponding age of ES would then be 4561 Ma. We note, however, that the initial ^{53}Cr/^{52}Cr in ES is close to 0 ε and that the published ^{54}Cr/^{52}Cr ratio is anomalously high: ~+1.5 ε. A very similar isotope signature was found for the bulk Allende sample (^{53}Cr/^{52}Cr $= 0.22 \pm 0.22$ ε; ^{54}Cr/^{52}Cr $= 1.8 \pm 0.4$ ε) by Birck and Allègre (1988). This, combined with the almost perfect match between the oxygen isotopic composition in bulk CV3 chondrites and that in ES (Clayton and Mayeda, 1996), may indicate that CV3 type material could serve as a precursor of the ES parent body. Carbonaceous chondrites, however, bear a presolar Cr component which could influence the apparent Mn-Cr isotope systematics (section 3.4). On the other hand, the less precise, though pioneering measurements of ES and Allende may also be less accurate and possible second order effects in the data may have escaped attention. The chronological meaning of the ES Mn-Cr age, therefore, has to be treated with caution.

Pallasites seem to be suitable objects to establish a chronological relationship between the relative abundances of ^{53}Mn and another short-lived radionuclide, ^{107}Pd ($T_{1/2} = 6.5$ Ma). For example, the ^{107}Pd/^{108}Pd ratio in the pallasites Brenham and Glorieta Mountain is ~1×10^{-5} (Chen and Wasserburg, 1996). Assuming that the ^{107}Pd-^{107}Ag isotope system closed in these pallasites roughly at the same time as did ^{53}Mn-^{53}Cr in OM and SPR then a ^{107}Pd/^{108}Pd ratio of ~1×10^{-5} would correspond to a ^{53}Mn/^{55}Mn ratio of ~1×10^{-6}. This semiquantitative estimate is, to a first approximation, consistent with the relative abundances of ^{107}Pd and ^{53}Mn in the IIIB iron meteorites (Chen and Wasserburg, 1996). Evidently, to obtain more precise constraints the ^{107}Pd-^{107}Ag and ^{53}Mn-^{53}Cr systems have to be studied in the same pallasite.

Acapulco. This primitive achondrite has a bulk chemical composition similar to that of H-chondrites but exhibits an igneous texture (Palme et al., 1981). The results for the study of the ^{53}Mn-^{53}Cr system were published by Zipfel et al. (1996). The data yield a ^{53}Mn/^{55}Mn ratio of $(7.5 \pm 1.4) \times 10^{-7}$ at the time of isotopic closure. The ^{53}Cr excess corresponding to a chondritic ^{55}Mn/^{52}Cr ratio obtained from the ACA isochron is ~0.56 ε. This ^{53}Cr excess is indistinguishable within the uncertainties from those of other meteorite classes (see above) and the ACA parent body is, thus, characterized by the same intrinsic ^{53}Cr/^{52}Cr. The calculated absolute age of ACA is then 4555.1 ± 1.2 Ma which is in good agreement with its Pb-Pb age of 4557 ± 2 Ma (Göpel et al., 1992).

Table 7 presents a summary of all new chronological information discussed in this article. It is hoped that the reader may thus have easier access to the data scattered throughout the text and figures.

3.2.5. Radial gradient in relative ^{53}Cr abundances

A close agreement of the intrinsic ^{53}Cr/^{52}Cr ratios in various classes of meteorites indicates that the difference in original ^{53}Mn abundances among the asteroid belt objects studied so far is marginal, if it exists at all. Thus we conclude that the ^{53}Mn-^{53}Cr system can be used as a chronometer for dating samples which have formed within the asteroid belt. However, the radial ^{53}Cr gradient discussed above and by Lugmair et al. (1996) still exists. With the new HED data, however, it is now a linear function of the heliocentric distance: the data points for the Earth-Moon system, Mars, and Vesta form a perfectly straight line (Fig. 16). It has to be emphasized that the existence of the ^{53}Cr gradient in the solar system is based solely on experimental data and no assumptions or models are involved (other than the assignment of the SNC meteorites to Mars and the HED meteorites to Vesta). The differences in the relative abundance of ^{53}Cr do exist regardless of the choice of scenarios and possible explanations for their presence.

There are, of course, several questions which at present cannot be answered with a high degree of confidence. One of the most important questions is: what mechanism is responsible for the observed radial gradient of ^{53}Cr. It could be a process of an early Mn/Cr fractionation in the nebula with an originally homogeneous ^{53}Mn distribution or the ^{53}Cr gradient may be a result of an indigenous heterogeneous radial ^{53}Mn distribution. As one reviewer has pointed out there is the third possibility that the original ^{53}Cr/^{52}Cr ratio itself was heterogeneous. This would then have to be a residual heterogeneity from the molecular cloud preserved as the observed gradient. We consider this possibility the least likely and, thus, will not further discuss it. (The first two possibilities are discussed in detail in section 3.3). Also, it is not yet clear if the parent bodies of the studied

Table 7. Summary of the initial $^{53}Cr/^{52}Cr$ and $^{53}Mn/^{55}Mn$ ratios, the relative ^{53}Mn-^{53}Cr ages (ΔT), and absolute ages (T).

Sample[#]	$(^{53}Cr/^{52}Cr)_0$ (ϵ-units)	$^{53}Mn/^{55}Mn$ ($\times 10^6$)	ΔT (Ma)	T (Ma)
		Angrite		
LEW	0.40 ± 0.16	1.25 ± 0.07	$\equiv 0$	4557.8 ± 0.5*
		Eucrites		
CAL	1.12 ± 0.06	≤ 0.12	≤ -13	≤ 4545
CK	0.49 ± 0.09	3.7 ± 0.4	5.8 ± 0.8	4563.6 ± 0.9
EET	1.12 ± 0.06	≤ 0.25	≤ -9	≤ 4549
IB	0.79 ± 0.06	1.06 ± 0.50	-1^{+2}_{-4}	4557^{+2}_{-4}
JUV	0.56 ± 0.09	3.0 ± 0.5	4.7 ± 0.9	4562.5 ± 1.0
MC	0.88 ± 0.07	≤ 0.25	≤ -9	≤ 4549
POM	0.71 ± 0.07	≤ 0.6	≤ -4	≤ 4554
SM	0.56 ± 0.06	0.54 ± 0.23	-5^{+2}_{-4}	4553^{+2}_{-4}
		Diogenites		
JT	0.57 ± 0.05	≤ 2	≤ 3	≤ 4561
SHA	0.39 ± 0.07	≤ 1	≤ -1	≤ 4557
		HED PB**		
HED PB	0.25 ± 0.07	4.7 ± 0.5	7.1 ± 0.8	4564.8 ± 0.9
		Pallasites		
OM	0.42 ± 0.08	1.29 ± 0.19	0.2 ± 0.9	4558.0 ± 1.0
SPR	nd	~ 1	~ -1	~ 4557
		Acapulco		
ACA	0.51 ± 0.07	0.75 ± 0.14	-2.7 ± 1.1	4555.1 ± 1.2

[#] LEW86010 (LEW), Caldera (CAL), Chervony Kut (CK), EET87520 (EET), Ibitira (IB), Juvinas (JUV), Moore County (MC), Pomozdino (POM), Serra de Magé (SM), Johnstown (JT), Shalka (SHA), Omolon (OM), Springwater (SPR), Acapulco (ACA) (Zipfel et al., 1996).

* The absolute Pb-Pb age of LEW (Lugmair and Galer, 1992) is used as a reference value.

** HED PB, Howardite-Eucrite-Diogenite Parent Body.

meteorites all originated within a relatively narrow region of the asteroid belt with no detectable ^{53}Cr heterogeneity or if the whole asteroid belt is characterized by similar $^{53}Cr/^{52}Cr$ ratios. Although the data indicate a definite linear $^{53}Cr/^{52}Cr$ dependence on the heliocentric distance we actually do not know the form of this dependence outside the 1 AU - 2.4 AU range. It may become flat at distances of <1 AU if there was no ^{53}Mn addition to the very inner zones of the solar nebula and Earth has very little or no contribution of radiogenic ^{53}Cr or it may have a maximum at distances of >2.4 AU if, for example, the injection of the freshly synthesized material was very confined.

The extrapolation of the correlation line (Fig. 16) to zero AU yields $^{53}Cr/^{52}Cr$ of ~ -0.4 ε. This value may represent an initial $^{53}Cr/^{52}Cr$ ratio in the solar nebula in the case if the gradient is due to an original ^{53}Mn heterogeneity. If, however, the gradient is a result of an early Mn/Cr fractionation and the ^{53}Mn distribution was originally fairly homogeneous, the $^{53}Cr/^{52}Cr$ ratio at zero AU (that is in the Sun) cannot be estimated from these data.

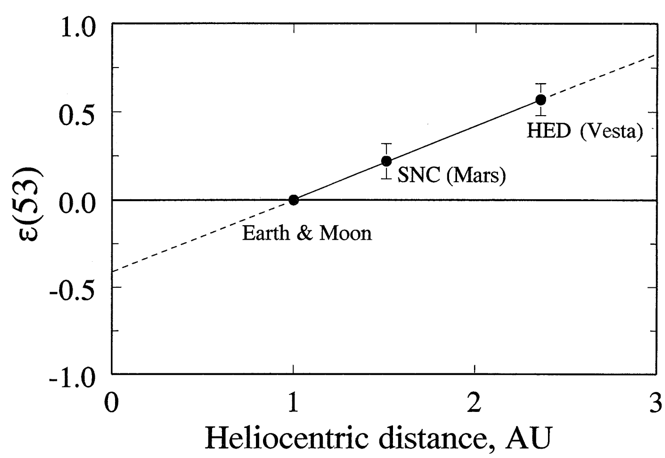

Fig. 16. Excesses of radiogenic ^{53}Cr in solar system bodies vs. their heliocentric distance. Experimental data indicate that the ^{53}Cr excesses in the earth-moon system, SNC PB (Mars), and HED PB (Vesta) is a linear function of the heliocentric distance. The extrapolation of the correlation line to a zero distance yields $^{53}Cr/^{52}Cr$ of ~ -0.42 ε which may represent a solar system initial $^{53}Cr/^{52}Cr$ if the observed radial gradient of $^{53}Cr/^{52}Cr$ is due to an original ^{53}Mn heterogeneity.

3.3. Indigenous ^{53}Mn Heterogeneity or Mn/Cr Fractionation?

3.3.1. *Early Mn/Cr fractionation*

It is well known that Mn is a moderately volatile element while Cr is somewhat more refractory. The condensation temperatures of Mn and Cr are 1190 K and 1277 K, respectively (Wasson, 1985). One can intuitively assume that there was a temperature gradient in the early solar system: the inner regions which were closer to the sun could maintain higher temperatures than the outer regions. This assumption is consistent with the observation that asteroid compositions are heliocentrically zoned: objects which are believed to have been exposed to high temperatures, causing their metamorphism or even melting, are located closer to the sun while those unaltered or only mildly altered are located farther outside (Bell et al., 1989). The heating might be caused by a high radiative stage of the sun or, if several Ma time scales for the accretion of asteroidal size objects is accepted, by the decay of ^{26}Al (Lee et al., 1976; Grimm and McSween, 1993). This might produce a Mn/Cr fractionation due to differences in volatilities and might result in a radial depletion of Mn relative to Cr. If this hypothetical fractionation occurred during the very early stages of the solar system this could produce the observed radial gradient in the radiogenic ^{53}Cr abundances. It is also well known that all objects in the solar system show different degrees of volatile depletion as compared to CI carbonaceous chondrites. However, the processes which governed these depletions are poorly constrained. It is difficult to specify definite conditions for the loss of volatiles, and there is no explicit evidence that decreasing depletion of moderately volatile elements in general, and Mn in particular, reflect formation at increasing heliocentric distances. As was noted by Palme et al. (1988), the lack of meteorites enriched in moderately volatile elements makes it unlikely that the missing volatiles were transported outwards, away from the Sun. In addition, midplane temperatures calculated by modeling of planet formation (Tscharnuter and Boss, 1993) indicate a flat profile in the inner nebula (~1500 K) at distances of ≤4 AU. Nevertheless, the possibility of radial Mn/Cr fractionation may still be plausible and has to be considered.

Studies of the Earth's composition (e.g., Dreibus and Wänke, 1979, 1980; Wänke, 1981) have shown than Mn is preferentially depleted in the terrestrial mantle as compared to Cr and is residing in the core. In contrast according Drake et al. (1989), the experimentally obtained distribution coefficients of Cr and Mn between solid metal, liquid metal and liquid silicate suggest that the abundances of Cr and Mn do not reflect core formation in the Earth. Therefore, the Mn depletion in the Earth would have to be the result of its volatility and the mantle ^{55}Mn/^{52}Cr ratio of 0.40 ± 0.04 (Drake et al., 1989), most likely, represents the bulk Earth. This ratio is significantly lower than that in chondrites (0.76). In this scenario one could then assume that the lower ^{53}Cr/^{52}Cr in the Earth is due to the lower Mn/Cr ratio. Obviously, because of the short half life of ^{53}Mn, the process of Mn depletion must have occurred not during the accretion stage of the Earth but during that of the earth's precursors or during the nebular stage of the solar system. From the terrestrial and chondritic ^{55}Mn/^{52}Cr and ^{53}Cr/^{52}Cr ratios one can estimate a ^{53}Mn/^{55}Mn ratio of $\sim 1.5 \times 10^{-5}$ at the time of Mn depletion. This would correspond to a time of ~4571 Ma ago. This time predates the formation of Allende CAIs (4566 ± 2 Ma, Göpel et al., 1991) by at least 3 Ma and, therefore, suggests an early nebula process. On the other hand, the experimental results of Goarant et al. (1992) have shown that, in contrast to moderate pressures, at a high pressures (70 GPa) MnO and Cr_2O_3 are very soluble in molten iron. Allègre et al. (1995) calculated Mn and Cr concentrations in the Earth's core of 5820 ppm and 7790 ppm, respectively. This yields a bulk Earth ^{55}Mn/^{52}Cr ratio of 0.68, which is much higher than that of Drake et al. (1989) and relatively close to chondritic (0.76). Furthermore, in contrast to Drake et al. (1989), in a new model of Wänke and Dreibus (1997) the inferred concentrations of Mn and Cr in the core are 0.7% and 0.68% and the Mn/Cr ratio of the bulk Earth is chondritic.

A moderate depletion of Mn relative to Cr in the bulk Earth, corresponding to a bulk ^{55}Mn/^{52}Cr ratio of 0.68 (Allègre et al., 1995), does not allow to explain the lower ^{53}Cr/^{52}Cr ratio in the Earth by an early Mn/Cr fractionation. The ^{53}Mn/^{55}Mn ratio at the time of this event, estimated from the terrestrial ^{55}Mn/^{52}Cr ratio of 0.68 and the chondritic ^{53}Cr/^{52}Cr and Mn/Cr ratios, would be $\sim 7 \times 10^{-5}$ and would correspond to 4579 Ma ago, 13 Ma before the formation of CAI's.

Although the first scenario would yield a more reasonable timescale, both appear rather implausible when Mars is considered. The model Mn/Cr ratio of Mars, although estimated with a large uncertainty, is chondritic (Dreibus and Wänke, 1979, 1980; Drake et al., 1989), and no Mn depletion has ever been postulated for Mars. However, the ^{53}Cr excesses in SNC meteorites are clearly only about half of those in chondrites and $\sim 0.2\ \varepsilon$ higher than the terrestrial value. A clear solution to and elimination of the volatility problem would come from the direct measurement of a meteorite type with a significantly lower or higher than chondritic ^{53}Cr excess but with a chondritic Mn/Cr ratio.

3.3.2. *^{53}Mn heterogeneity*

Although the possibility of an early Mn/Cr fractionation cannot be totally excluded at present, the results of a simple estimate favor a different mechanism for the presence of the radial gradient in ^{53}Cr abundances. Let us assume that ^{53}Mn was originally homogeneously distributed in the solar system and the observed ^{53}Cr heterogeneity is then due to a Mn/Cr fractionation. The Pb-Pb ages of the angrites (4557.8 ± 0.5 Ma, Lugmair and Galer, 1992) and CAI's, believed to be among the first solar system condensates ($T_{CAI} = 4566 \pm 2$ Ma, Göpel et al., 1991; see also Tatsumoto et al., 1976; Chen and Tilton, 1976; Chen and Wasserburg, 1981), yield the maximum possible time difference between their formation (ΔT) of 10.7 Ma. From the angrite ^{53}Mn/^{55}Mn ratio of 1.25×10^{-6} and $\Delta T = 10.7$ Ma we calculate a ^{53}Mn/^{55}Mn ratio at time T_{CAI} of 9.2×10^{-6}. Since we assumed a homogeneous ^{53}Mn distribution within the solar system this ratio should basically represent the initial value for all solar system objects, in particular, in chondrites and in the Earth's precursors.

Because the present day ^{53}Cr/^{52}Cr ratio is determined by

$$^{53}\mathrm{Cr}/^{52}\mathrm{Cr} = (^{53}\mathrm{Cr}/^{52}\mathrm{Cr})_0 + (^{55}\mathrm{Mn}/^{52}\mathrm{Cr})\,(^{53}\mathrm{Mn}/^{55}\mathrm{Mn})_0, \tag{2}$$

where subscript 0 indicates the initial ratios, we calculate from the chondrite data ($^{53}Cr/^{52}Cr = 0.48\ \varepsilon$, $^{55}Mn/^{52}Cr = 0.76$) the initial $(^{53}Cr/^{52}Cr)_0$ in the solar system of $-0.14\ \varepsilon$. Using this initial and the present day $^{53}Cr/^{52}Cr$ for the Earth of 0 ε we calculate a Mn/Cr ratio for the Earth of 0.15 which is unreasonably low. A similar estimate for Mars yields a Mn/Cr ratio for this planet of 0.4 while the inferred value is almost twice as large: 0.71 (Drake et al., 1989). Therefore, the assumption of homogeneous ^{53}Mn distribution is most likely incorrect. Of course, this estimate is only valid if the Pb-Pb ages of CAI's do reflect the true time of their formation rather than a late reequilibration of the U-Pb system at this time. As will be shown below, the Mn-Cr isotope data are still consistent with the upper limit for the Pb-Pb age of CAI's of 4568 Ma.

What, then, was the physical process which caused the heterogeneous distribution of ^{53}Mn in the late solar nebula? If a cloud of ejecta from a nearby star, which contains freshly synthesized elements including ^{53}Mn (and other short-lived radionuclides), falls onto the passive disk and the fresh material does not mix thoroughly with the whole solar system matter, its location may be somewhat confined to regions where it was originally injected. Such an incomplete mixing might result in the radial gradient of the ^{53}Mn distribution which is now frozen into inner solar system bodies as characteristic and variable $^{53}Cr/^{52}Cr$. This simple scenario, however, is not supported by models implying a thorough mixing in radial directions (e.g., Wetherill and Chambers, 1997; Shu et al., 1996). The ^{53}Mn gradient might also be generated directly within the solar nebula. After the completion of disk accretion and of accretion of matter onto the central Sun a T Tauri Sun emits energetic protons and alpha particles that strike the surface of the quiescent disk where they are stopped within their range (Clayton and Jin, 1995). ^{26}Al, ^{53}Mn, and the other short-lived radionuclides are thus formed in nuclear reactions within the disk skin. Variations of flux and density conditions at different heliocentric distances may then give rise to a heterogeneous ^{53}Mn distribution, including that of other freshly produced nuclei. In addition, the relative volatility of the freshly produced nuclei could contribute to the establishment of a gradient. Still, relative production rates (e.g., of ^{26}Al and ^{53}Mn) are inconsistent with observed initial abundances of the short lived nuclei. Problems of similar nature also exist with the model of Shu et al. (1996). Moreover and as mentioned above, any true stochastic mixing mechanism at the planetary embryo stage as, for example, favored by Wetherill and coworkers, would totally eradicate any previously existing gradient. It is, therefore, clear that none of the conditions set by existing scenarios can satisfactorily account for the presence of the $^{53}Cr/^{52}Cr$ gradient among inner solar system objects. This gradient, however, is an observational and thus model independent fact and has to be taken as an inevitable constraint in future calculations.

3.4. Age Of The Solar System

The oldest high-precision ages among solar system objects were obtained for CAI's from the Allende CV3 chondrite. A model Pb-Pb age of 4566 ± 2 Ma (Göpel et al., 1991) provides the best estimate for the first condensation of matter in the solar system. However, if this average CAI age reflects processes of late reequilibration rather than the true crystallization age, which cannot be excluded, this value would represent just a lower limit for the time of CAI formation. The ^{53}Mn-^{53}Cr systematics enable us to obtain additional constraints for the solar system age. In the following we imply that the onset of ^{53}Mn decay within solar system matter was concurrent with the formation of the solar system. This is true if ^{53}Mn and other short lived nuclei were produced in stars and were injected into the proto-solar cloud. If, on the other hand, these nuclei were produced within the solar nebula by solar radiation then the following estimates date these processes which, by some models (e.g., Shu et al., 1996), would also coincide with the formation of CAI.

In section 3.2.3 we have shown that the bulk HED PB Mn/Cr ratio is essentially chondritic, that the last global Mn/Cr fractionation in the HED PB mantle occurred 4564.8 Ma ago and that the initial $^{53}Cr/^{52}Cr$ ratio at this time was ~0.25 ε. Using these data the lower limit for the solar system age can be deduced. Indeed, it is clear that the solar system initial $^{53}Cr/^{52}Cr$ (SSI) cannot be higher than the terrestrial $^{53}Cr/^{52}Cr$. Assuming the terrestrial $^{53}Cr/^{52}Cr$ (0ε) = SSI we calculate T_0-T_1 (Fig. 14), that is, the time required to evolve $^{53}Cr/^{52}Cr$ in a chondritic source from 0 ε to 0.25 ε. Using Eqn. 3

$$^{53}Cr/^{52}Cr(T_1) = {}^{53}Cr/^{52}Cr(T_0) + {}^{55}Mn/^{52}Cr \times {}^{53}Mn/^{55}Mn(T_1) \times \{\exp[\lambda(T_0 - T_1)] - 1\}, \quad (3)$$

this time is ~3.2 Ma. Therefore, the minimum age of the solar system is ~4568.0 Ma. This value agrees with the upper limit for the absolute age of CAI's of 4566 ± 2.0 Ma. A similar calculation using the same initial and the present day chondritic $^{53}Cr/^{52}Cr$ ratio of 0.48ε and the $^{55}Mn/^{52}Cr$ ratio of 0.76 yields a minimum solar system age of ~4567.1 Ma. It has to be noted that this simple calculation does not require any assumption on the mechanism causing the ^{53}Cr heterogeneity.

An upper limit for the solar system age can also be obtained. We have shown in section 3.2.5 that the extrapolation of the correlation line in Fig. 16 to zero heliocentric distance yields $^{53}Cr/^{52}Cr$ of ~−0.42 ε. This value may represent an initial $^{53}Cr/^{52}Cr$ ratio in the solar nebula for the case where the gradient in ^{53}Cr abundances is due to an original ^{53}Mn heterogeneity. Then, using SSI = -0.42ε, we obtain for the evolution of $^{53}Cr/^{52}Cr$ to the HED initial value of ~0.25 ε a time difference T_0-T_1 of ~6.2 Ma. This yields an upper limit for the solar system age of ~4571 Ma. This calculation is based on the assumption of an original ^{53}Mn heterogeneity and a chondritic Mn/Cr ratio. In another way, we can calculate the solar system initial $^{53}Mn/^{55}Mn$ ratio from the observed $^{53}Cr/^{52}Cr$ ratio in angrites (~0.48 ε; this is the same as in chondrites) and SSI. This yields a $(^{53}Mn/^{55}Mn)_I = 1.34 \times 10^{-5}$. We then can calculate that it takes ~12.7 Ma for ^{53}Mn to decay from the initial value to that found in angrites (1.25×10^{-6}). The sum of this decay interval and the absolute age of the angrites (4557.8 Ma) again leads to a total age of ~4571 Ma.

As mentioned above, an interesting result, although this may be coincidental, is obtained if we instead assume that the lower $^{53}Cr/^{52}Cr$ ratio in the Earth is due to an early global Mn/Cr fractionation in the nebula. If we use a bulk Earth $^{55}Mn/^{52}Cr$ ratio of 0.40, as was suggested by Drake et al. (1989), we get the same value of ~4571 Ma. Thus, regardless of the scenario, the ^{53}Mn-^{53}Cr systematics indicates rather narrow limits for the age of the solar system: 4568–4571 Ma. Combining the time limits obtained with ^{53}Mn-^{53}Cr systematics and those from the Pb-Pb data on CAIs of 4564–4568 Ma, a current best estimate for the solar system age is 4568 Ma. Obviously, this estimate implies that the time span between the onset of ^{53}Mn decay in the nebula and the formation of CAIs was very short. It also assumes that the CAI ages measured by Göpel et al. (1991) are true solidification ages and that no earlier generation of CAIs exists.

The age of 4568 Ma is also supported by the ^{26}Al data in feldspars from the H4 chondrite Ste. Marguerite (Zinner and Göpel, 1992). These data yield a $^{26}Al/^{27}Al$ ratio of $(2.0 \pm 0.6) \times 10^{-7}$. If interpreted chronologically and compared with the canonical $^{26}Al/^{27}Al$ ratio of 5×10^{-5} in CAIs, this value gives a time difference between the isotope closure of the ^{26}Al-^{26}Mg system in Ste. Marguerite and CAIs of 5.6 ± 0.4 Ma. Since the Pb-Pb age of Ste. Marguerite is 4562.7 ± 0.6 Ma we obtain for CAIs an age of 4568.3 ± 0.7 Ma.

Nyquist et al. (1997) have recently studied the ^{53}Mn-^{53}Cr systematics in bulk chondrules from the primitive chondrites Bishunpur (LL3.1) and Chainpur (LL3.4). The results yield $^{53}Mn/^{55}Mn$ ratios at the time of chondrule formation of $\sim 9 \times 10^{-6}$ (Nyquist, 1997, pers. commun.) which corresponds to a roughly contemporaneous formation of the chondrules and Allende CAIs. Taking into account the uncertainties in the slopes of the bulk chondrule isochrons a time span of $1-2$ Ma between formation of these objects is still allowed.

Several comments have to be made on the presently unresolved inconsistency between the time scales obtained from CAIs with the ^{53}Mn-^{53}Cr and Pb-Pb chronometers. The Pb-Pb data yield a time interval between formation of the angrite LEW86010 (Lugmair and Galer, 1992) and Allende CAIs (Göpel et al., 1991) of ~8 Ma while the time difference obtained from the $^{53}Mn/^{55}Mn$ ratios for LEW86010 and CAIs (Birck and Allègre, 1988) is as large as ~18 Ma. The comparison between $^{53}Mn/^{55}Mn$ ratios in CAIs of $\sim 4.4 \times 10^{-5}$ (Birck and Allègre, 1988) and in chondrules from the primitive chondrites Bishunpur (LL3.1) and Chainpur (LL3.4). of $\sim 9 \times 10^{-6}$ (Nyquist et al., 1997; pers. commun.) yields a time difference between their formation of ~8 Ma. This large time difference appears to be unlikely because of the astrophysical difficulties to store CAIs in the nebula for such a long time before the formation of chondrules without the CAIs falling into the sun due to drag-induced radial drift (Weidenschilling, 1977). If indeed true, a very high $^{53}Mn/^{55}Mn$ ratio for CAIs would reflect a grossly heterogeneous distribution of ^{53}Mn in the early solar system. Models for a local production of short-lived radionuclides and CAIs, as put forward by Shu et al. (1996) or in a thin outer layer of the passive disk (Clayton and Jin, 1995) offer a possible explanation for the enhanced $^{53}Mn/^{55}Mn$ ratios in CAIs. Russell et al. (1996), however, argued against such a model and concluded that the original abundances of the short-lived ^{26}Al were the same in CAIs and chondrule material. One can expect a similar behavior for ^{53}Mn if these two radionuclides were formed by a similar process and at the same time. While it is not yet clear whether the original ^{53}Mn abundances were similar in CAIs and in other solar system objects, there is at least unequivocal evidence that the Chromium isotopic composition in CAIs (normal or FUN) is anomalous (e.g., Papanastassiou, 1986) and could severely restrict the application of the ^{53}Mn-^{53}Cr system as a chronometer for CAIs.

The analysis of the data by Birck and Allègre (1988) indicates that there is an anti-correlation between the measured ^{53}Cr and ^{54}Cr excesses in the CAIs. This is also true for the Allende bulk leaches (Rotaru et al., 1989). These effects suggest the presence of a presolar Cr component with an anomalous chromium isotopic composition and may mean that the measured $^{53}Cr/^{52}Cr$ ratios in Allende CAIs reflect a complex superposition of the ^{53}Mn decay process (note: the nucleosynthetic production of ^{53}Cr is through ^{53}Mn) and mixing of at least two Cr components with anomalous isotopic composition. We believe, therefore, that the chronological meaning of the $^{53}Mn/^{55}Mn$ ratios in CAIs is still tentative. A presolar and unhomogenized Cr was also found in the other types of carbonaceous chondrites, e.g., in Cr-rich Murchison spinels (Esat and Ireland, 1988), and in the Alais, Felix, Karoonda, Murchison, Murray, and Orgueil leaches (Rotaru et al., 1989; Rotaru et al., 1990; Podosek et al., 1997). In some cases (Murchison, Murray, Orgueil; Rotaru et al., 1989) the anti-correlation between the ^{53}Cr and ^{54}Cr excesses can also be observed. For this reason, any conclusions on a chronological meaning of calculated $^{53}Mn/^{55}Mn$ ratios in carbonaceous chondrites should be viewed with caution.

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